



AfricaFertilizer.org

Training Manual on Fertilizer Statistics in Africa

*To feed our people
we must first
feed our soil.*

~ Olusegun Obasanjo

June 2012 – draft version.

AfricaFertilizer.org is a joint initiative from IFDC, FAO, IFA and the Africa Union to facilitate exchange of information about soil fertility, fertilizers and good agricultural practices in Africa



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List of Acronyms

AU	African Union
CIF	Cost Insurance & Freight
DAP	Diammonium Phosphate
DRC	Democratic Republic of Congo
FAO	Food and Agriculture Organisation
FRUA	Fertilizer Resource Utilization Account
FUBC	Fertilizer Utilization By Crop
IFA	International Fertilizer Industry Association
IFDC	International Fertilizer Development Centre
LC	Letter of credit
NEPAD	New Partnership for Africa Development
NGO	Non-Governmental Organization
NPK	Nitrogen, Phosphorus and Potassium (Fertilizers)
SC	Supply Chains
SSA	Sub Saharan Africa
HS	Harmonized Systems
WCO	World Customs Organization

Useful Websites

AfricaFertilizer.org	www.africafertilizer.org
AMITSA	www.amitsa.org
Africa Union	www.africa-union.org
CountrySTAT	www.countrystat.org
FAO (ESS)	www.fao.org/economic/ess/ess-home/en/
FAO STAT	http://faostat.fao.org/site/575/default.aspx#ancor
IFA	www.fertilizer.org
IFDC	www.ifdc.org
NEPAD	www.nepad.org

Website:	www.africafertilizer.org
Facebook:	www.facebook.com/africafertilizer.org
Twitter:	@AfricFertilizer
Email:	info@africafertilizer.org

Introduction

In June 2006, Africa's leaders gathered in Abuja for the Africa Fertilizer Summit, with an aim to highlight the importance of fertilizers in achieving an African Green Revolution. The principal outcome of the Summit, the ***Abuja Declaration on Fertilizers for an African Green Revolution*** confirms the commitment of the African head of states to achieving a rapid increase in fertilizer use on the continent, from the current average of 9 kilograms per hectare to an average of at least 50 kilograms per hectare by 2015. The Declaration delineates concrete measures and actions to rapidly accelerate the accessibility, availability and affordability of fertilizers on the continent.

These actions can be summarized as follows:

1. harmonize policies and regulations to ensure duty- and tax-free movement of fertilizers across regions, and the development of capacity for quality control;
2. develop and scale up agro dealer networks to improve farmer access to fertilizer;
3. develop and strengthen the capacity of the privates sector involved in agriculture;
4. grant targeted subsidies, with special attention to poor farmers;
5. improve output market incentives, particularly by accelerating investment in infrastructure and strengthening farmers' organizations ;
6. establish national financing facilities for input suppliers to accelerate access to credit;
7. promote the establishment of regional fertilizer procurement and distribution facilities through strategic public-private partnerships;
8. promote national/regional fertilizer production and intra-regional fertilizer trade;
9. improve farmer access to complementary inputs (quality seeds, irrigation facilities, extension services, market information, and soil nutrient testing and mapping;
10. establish an Africa Fertilizer Development Financing Mechanism.

Hence, the genesis of the AfricaFertilizer.org portal is the Abuja Declaration which confirms the lack of market information as an important constraint to fertilizer market development on the continent. Accordingly, in 2009 IFDC initiated the development of the AfricaFertilizer.org (AFO) portal. The aim of the AFO is to provide a unique combination of information and data on fertilizers and soil fertility issues in Africa. It does this by sourcing, aggregating, filtering and sharing information on fertilizer from and to international, regional and national players in the sector. It also engages the small, local fertilizer dealers and agricultural extension workers who are the "last mile" link with millions of smallholder farmers. Therefore, the Africafertilizer.org portal includes a web-based data center that has been designed to manage the following data and information: fertilizer production, fertilizer imports and exports, fertilizer production capacity, fertilizer world prices, national fertilizer retail prices, total fertilizer consumption, fertilizer use by crop and fertilizer market intelligence.

Agreement has been reached for the website to be established as a consortium, led by IFDC, supported by a short list of key partners (AU/NEPAD Agency, FAO and its "CountryStat for Sub-Saharan Africa" project, and the International Fertilizer Industry Association (IFA)). The website is currently hosted and managed by IFDC.

CountrySTAT – a web-based information system - is a project partnership between FAO and the Gates Foundation to substantially improve the quality, accessibility, relevance, and reliability of national statistics on food and agriculture in 17 Sub-Saharan African countries. Through national and regional CountrySTAT projects, FAO forms partnerships with statistical offices and the relevant ministries to introduce the system and build the national capacity to use it. FAO and its CountrySTAT project will partner with IFDC, IFA and NEPAD through technical collaboration in a number of areas, including contribution of information, data and statistics related to fertilizers in Africa and sharing and improving methodologies, data management expertise and related tools. IFA and FAO collect data on African fertilizer statistics, but to date, [high quality data can be found only in] North Africa and South Africa. The key contribution by the AFO consortium will be to improve data from the rest of Africa for the website.

At the request of the AU Heads of State, NEPAD has been monitoring and reporting on progress in the implementation of the Abuja Declaration on Fertilizers since June 2006. Accordingly, NEPAD's contribution to the partnership will be to provide information and data on fertilizer market intelligence in SSA, especially countries affiliated to IFDC.

Based on their experience to date in collecting, analyzing and disseminating statistics and information on fertilizer markets in Africa, the partner organizations realize there is a need to: a) improve the availability and quality of fertilizer statistics at the national level; b) Improve the knowledge on the overall performance of fertilizer markets in Africa among; c) Improve the speed and frequency of dissemination of fertilizer statistics and information to ensure relevance to policy-making and program planning and implementation.

The purpose of this manual is to be used by CountrySTAT Trainers whose mandate is to train a larger number of national Trainees. Accordingly, **this training manual is designed to be used by AFO Trainers** who can provide training at the country level to:

1. Improve the collection of fertilizer statistics by national statistics officers in the CountrySTAT programs **(the “hard” data)**;
2. Improve the collection of fertilizer data and market intelligence by staff in the Ministry of Agriculture in charge of farm inputs **(the “soft” data)**.

The target group will be CountrySTAT trainers from key AfricaFertilizer.org regions and countries to provide training on:

- AfricaFertilizer.org initiative and partners
- Basic fertilizer technical knowledge (definitions, terminology, classifications, sources, key issues for data collection, questionnaires)
- Fertilizer soft and hard data and statistics (from collection to validation and publishing)

The manual is organized as follows:

Chapter 1 provides an overview of the fertilizer market in Africa, describes the key components of well-functioning fertilizer markets, and summarizes the current status of fertilizer markets in Africa vis-à-vis these components. The purpose is to provide trainees with sufficient grounding on the fertilizer markets in Africa to understand the importance of the data and information that will be collected and its role and relevance for improving the performance of the fertilizer markets in Africa.

Chapter 2 will present fertilizer, definitions, classifications, and conversion into nutrients.

Chapter 3 will present and describe the following variables: Fertilizer Production, (Trade) Fertilizer Imports, Fertilizer Exports, Fertilizer Consumption (Apparent or Actual), Fertilizer use (by crop, by acreage), and Prices. Collectively, the quantitative information on these variables will be known as **hard data**. It is very crucial for agricultural statistical officers to be conversant with the understanding of the above variables in order to facilitate data collection and analysis.

Chapter 4 will discuss **soft data issues**. Soft data complements hard data. It (soft data) concerns itself with issues around Fertilizer Policy and Network Building in the fertilizer industry in order to share information and risks.

Chapter 5 affords participants with the opportunity of applying what they have learnt in the training sessions through case studies.

Chapter 1. Fertilizer Sub-Sector in Africa – An Overview



Learning Objectives

- Participants are expected to appreciate the importance of fertilizer in agriculture
- To underscore the importance of local production of fertilizers as overreliance on imports may expose sub Saharan African countries to supply-shocks.
- Understanding of the Fertilizer Market in sub Saharan Africa
- Exploring ways in which SSA countries may be encouraged to make use of High-Analysis fertilizers
- Understanding the Demand-side and Supply-side constraints in fertilizer acquisition and use in SSA countries

Substantive use of fertilizer is a prerequisite for the success of efforts to improve agricultural productivity and hence increase agricultural output and farm income in Africa. No country in the world has achieved substantive agricultural growth without the use of fertilizers. But fertilizer use rates in Africa are low. Average fertilizer use in Africa is 10 kg/ha, equivalent to 10 % of the world average, and almost 20 times less than the average for Asia (191 kg/ha) and nine times less than the average for Latin America (94 kg/ha). Low fertilizer use per hectare does not auger well for achieving food security or reversing the severe soil nutrient depletion in Africa. Declining nutrient consumption and low fertilizer application rates have translated into cereal crop yields per hectare for Africa and SSA that are much lower than the world average (3.26 tons per hectare) and lower than what farmers are capable of achieving under conducive conditions. Low fertilizer use rates per hectare translate into small fertilizer markets and hence, a small share of global fertilizer consumption for Africa as a whole and for SSA in particular. Out of the 44 countries in sub-Saharan Africa, only nine countries use more than 50,000 nutrient tons and 10 countries use less than 10,000 nutrient tons.

1.1 African Fertilizer Markets – The Global Perspective

1.1.1 Consumption

Globally, fertilizer consumption in sub-Saharan Africa (SSA) was 1 million metric tons per year in 2005/06 and this level has not changed much up to 2011. Egypt, South Africa and Morocco are the leading fertilizer consuming countries in Africa as a whole; they consumed 3.2 million metric tons (mt) of nutrients and accounted for 69% of Africa's total fertilizer consumption in 2005/06. In global terms, global fertilizer nutrient consumption for Africa accounted for only 3.4 % in the period 2007-2011. The table below shows the consumption shares of different Fertilizer nutrients for different regions in the world.

Table 1: Regional & sub regional fertilizer consumption 2007/08-2011/12 (in %)

Regions & Subregions	N		P		K	
	Share of World Cons	Annual Growth	Share of World Cons	Annual Growth	Share of World Cons	Annual Growth
World		1.4		2.0		2.4
Africa	3.4	2.9	2.5	1.0	1.6	2.0
North America	13.5	0.3	12.0	0.5	17.1	0.7
Latin America	6.3	2.4	13.0	2.8	17.5	2.9
West Asia	3.5	1.7	3.3	1.0	1.4	2.4
South Asia	19.6	2.2	20.5	3.5	10.9	4.2
East Asia	38.3	1.3	36.1	1.9	35.2	3.3
Central Europe	2.7	1.8	1.5	1.2	2.4	1.0
West Europe	8.4	-0.3	5.6	-0.7	9.5	0.0
Eastern Europe & Central Asia	3.0	2.4	2.0	4.5	3.1	1.6
Oceania	1.4	4.9	3.5	1.7	1.3	2.1

Source: World Bank

1.1.2 Production

Fertilizer production in Africa is concentrated in North Africa and South Africa. Production in SSA is extremely small although there is some production starting up in Nigeria and forecasted for Tanzania and the Democratic Republic of Congo (DRC). A key reason for the lack of production is that SSA is largely deficient in the natural gas and raw material resources (phosphate rock, sulfur and potassium salts) required for fertilizer production. Although there are numerous small deposits of phosphate rock throughout SSA, substantial commercial deposits are found only in Tanzania, Togo and Senegal. The DRC is the only country in SSA with commercial deposits of potash. Nigeria, Angola, Equatorial Guinea, Ethiopia, Mozambique, Namibia, DRC, Madagascar and Tanzania are the only countries with natural gas. In comparison, there are significant phosphate rock deposits in South Africa and the country produces 90 % of its phosphate fertilizer requirements and imports the rest. It does not produce potassium fertilizers. In North Africa, Morocco, Tunisia, Algeria and Egypt have substantial resources of phosphate rock and natural gas and produce both phosphate and nitrogen fertilizers.

Total nutrient production in Africa was 5.5 million mt in 2005/06, an increase of 13 % from the 4.9 million mt produced in 1995/96. However, this gain was offset by a 75 % decrease in production in SSA – from 0.3 million mt to 0.09 million mt – during the same period. In terms of Africa's share of global fertilizer production, in 2005/06 the African fertilizer industry produced 3 % of the world's total production of 164.9 million, which is similar to its share 10 years previously in 1995/96 (Figure 4). Egypt, Morocco and Tunisia accounted for the majority (78 %) of the total production of 4.4 million mt. SSA accounted for only 1.8 % of global fertilizer production in 2005/06 (Table 3).

1.1.3 Trade

With regards to trade, trade of total nutrients (imports and exports) increased between 1995/96 and 2005/06 for Africa as a whole and for SSA. Total imports for Africa increased from 1.4 million mt of nutrients to 2.2 million mt of nutrients, an increase of 57 %. The increase for SSA was 25 %, from 0.8 million mt to 1 million mt. Total exports from Africa increased from 2.7 million mt to 3.1 million mt (an increase of 16 %). The increase for SSA was 6 %, from 78,000 mt to 82,500 mt. Therefore, Africa as whole is a net exporter; it exports more fertilizer than it imports. However, **SSA is a net importer of fertilizers.**

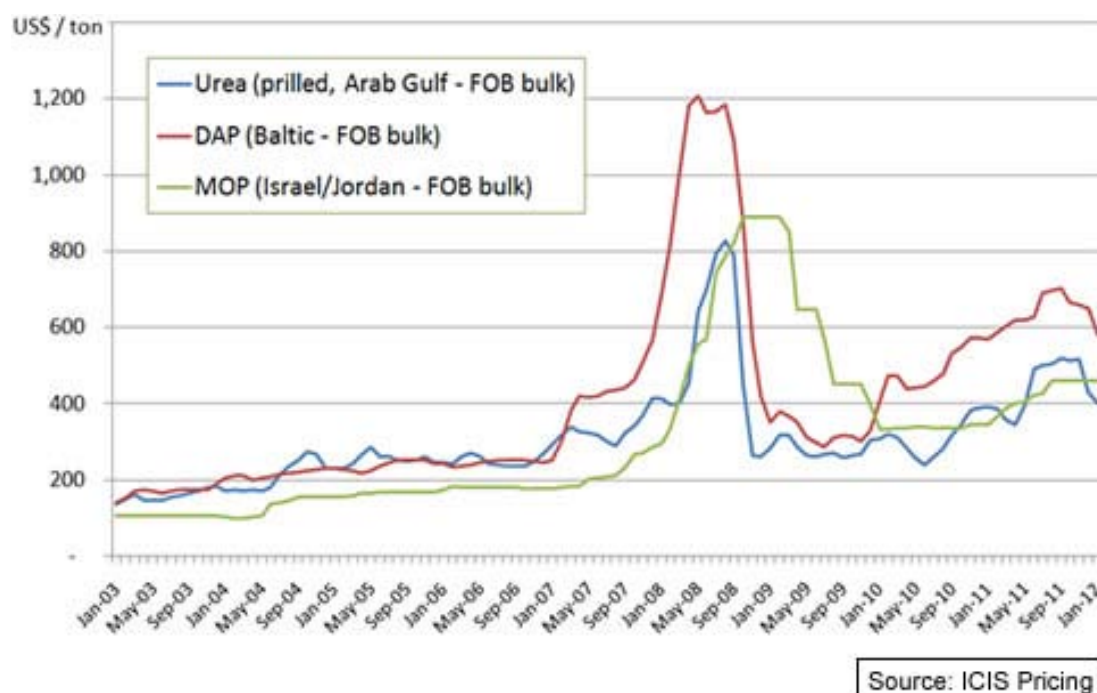
SSA's almost exclusive reliance on imported fertilizer for its supply was not an inherent disadvantage before 2007 since there was ample supply of fertilizers in the global fertilizer market at relatively stable prices. However, the unprecedented increase in fertilizer prices between January 2007 and mid-2008 was of extreme concern to African governments, and renewed interest in exploring the viability of national and regional fertilizer production in Africa. The US Gulf price of diammonium phosphate (DAP), a popular basal fertilizer for the production of maize in many African countries, increased by approximately 365 % from \$262/mt in January 2007 to \$1,218/mt by April 2008. Similarly, between January 2007 and August 2008, the price of urea in the Arab Gulf (urea is a popular top dressing fertilizer for maize production in Africa) increased by 200 %, from \$272/mt to \$815/mt.

These unprecedented fertilizer price increases were due to a number of factors, including oil prices at over \$100 per barrel that increased the demand for fertilizers to produce grain-based substitutes such as biofuel in Europe, the United States and Brazil; an increase in energy and freight prices; increased demand for grain-fed meat from the expanding middle-class societies of China, India and Brazil; and an increase in the price of natural gas, a major input for the production of ammonia for nitrogen fertilizers.

However, farmers could not afford to purchase fertilizers at these astronomical prices, resulting in a drop in fertilizer quantity demand, leading to stock accumulation and drastic price declines (by mid-December 2008 the price of DAP had fallen to \$469/mt and the price of urea was lower than pre-2007 levels at \$247/mt).

Although prices have since returned to pre-2007 levels due to the market correction, this experience clarified for African governments how vulnerably the continent is to global price volatility and has solidified interest in pursuing fertilizer production in Africa.

Figure 1: international prices for selected fertilizers (2003-2012)



Nevertheless, investment in fertilizer production is only economically viable if it is for production of 300,000 to 500,000 mt, which requires a capital investment of over \$500 million. Considering that only three countries in sub-Saharan Africa consume more than 100,000 mt of nutrients annually, investing in large-scale production facilities in individual countries is not feasible. One option to pursue in the long-run is to invest in fertilizer production in one country for local and regional consumption and export markets. However, for the short-term Africa will continue to import the majority of its fertilizer from the world market.

Therefore, the retail price of fertilizer in Africa is determined by three factors: the world market price; exchange rate; and domestic marketing costs. Since Africa is a consumer in the global market its response options in regard to reducing the price paid on the world market are limited. In this era of free trade and globalization, governments are also limited in regard to the extent by which they can manipulate the exchange rate to mitigate the impact of these price increases on farmers. Consequently, the key policy response has been a surge in the number of countries with subsidy programs. However, this response is expensive and in many cases unsustainable, and more importantly, it is typically implemented in a manner that eschews and undermines the private sector due to market distortions. A more viable alternative is to develop sustainable and competitive private-sector led fertilizer markets that will make available to farmers the appropriate fertilizers of high quality, at affordable prices, in a timely manner.

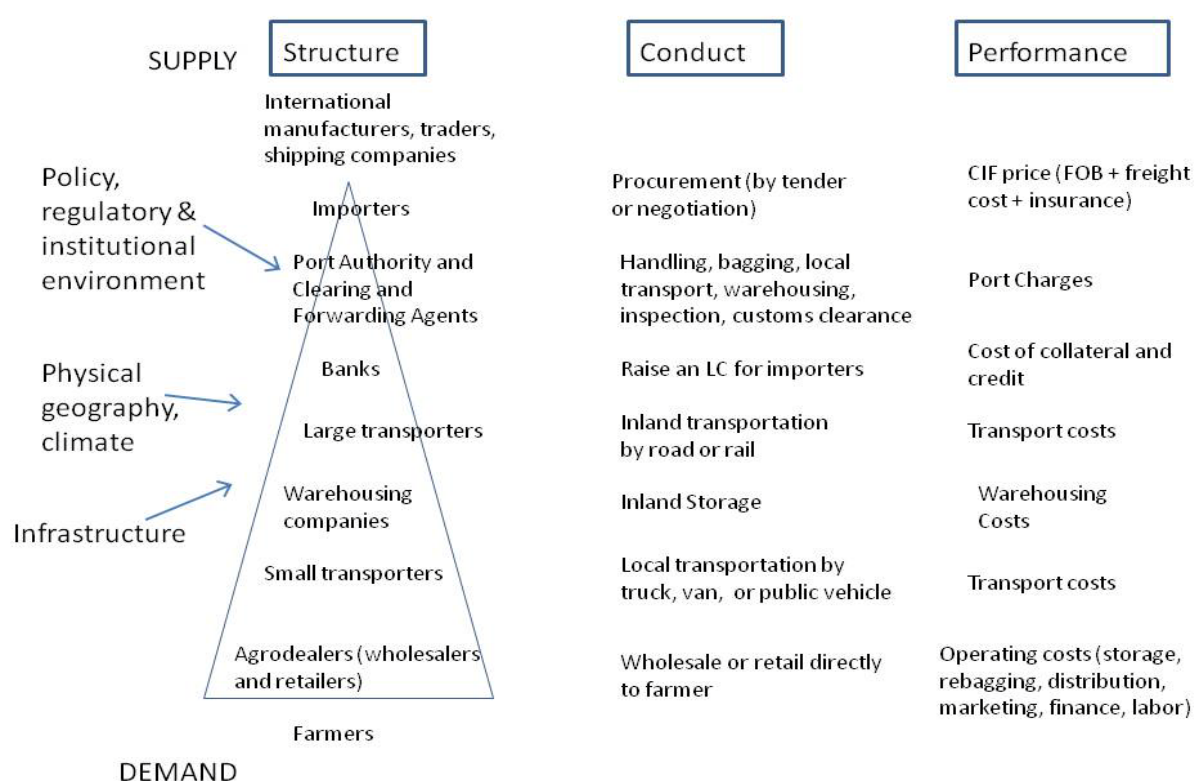
1.2 Functioning of Fertilizer Markets in Africa

Fertilizer markets refer to buying and selling, the supply chain from the point of production/importation through distribution to the final consumer (the farmer). Figure 1 illustrates

the physical flow of fertilizer from the overseas supplier to the farm gate in the typical fertilizer market in Africa. In doing so it depicts the players in the African fertilizer market (structure); the functions performed by these main actors (conduct) and the costs and margins generated as these functions are carried out (performance).

These functions are carried out within an enabling environment which is comprised of public policies, the regulatory framework, laws and institutions, the physical geography and climate of the country, and the infrastructural set-up. The magnitude of the costs and margins generated by carrying out each function is influenced by the policy, regulatory and institutional environment as well as the basic market conditions (size of the market, infrastructure) and physical conditions (climate, soils, land formations, etc.).

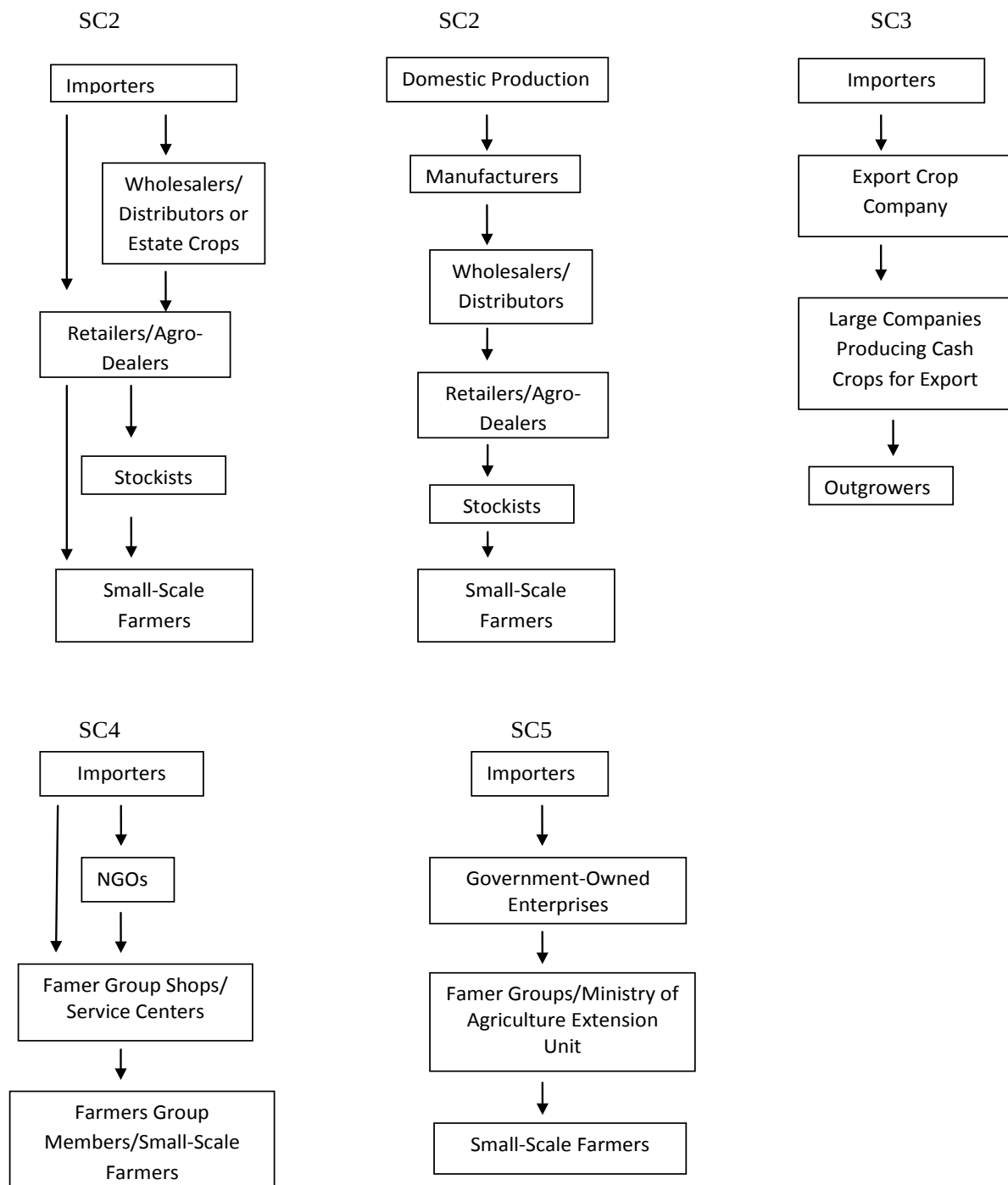
Figure 2: The Fertilizer Market in Africa



1.2.1 5 main types of supply chains

The fertilizer market in Africa is comprised of various types of **supply chains** (SC) with different numbers of stages and actors at each stage. (Figure 3)

Figure 3: Types of Fertilizer Supply Chains in SSA



SC1 represents the typical organization of fertilizer supply chains in Africa where markets have been liberalized. Importers procure fertilizer from overseas suppliers and sell it mainly to wholesalers/distributors but also directly to retailers/agro-dealers. The latter typically occurs in countries like Uganda where the wholesaler/distributor level does not exist or is insubstantial. Retailers/agro-dealers also sell fertilizers directly to small-scale farmers or sell it to stockists who then

sell it to small-scale farmers. However, these supply chains are typically weak and undeveloped due to a number of supply- and demand-side constraints which will be elaborated on below.

SC2 represents a more mature organization of fertilizer market with domestic production of fertilizers and distribution via well-developed wholesaler/distributor and retail networks. However, such systems only exist in North Africa and South Africa; there are few such systems in sub-Saharan Africa.

SC3 is commonly found in the export cash crop sectors of SSA such as tea and sugar in Kenya, tobacco in Malawi, cotton in West Africa, and sugarcane in Mauritius. These companies either procure fertilizer directly from overseas suppliers or place orders with local importers for their contracted outgrowers. They supply the fertilizer on credit and deduct the cost of the inputs and other services at harvest time.

SC4 is found where NGOs implementing development projects also procure fertilizers for their farmers via local importers as part of their project objective of improving food security. The NGOs provide the fertilizers to their farmers directly for free, at subsidized prices or on credit. Alternatively, NGOs help their farmer groups set-up farmer service centers which then purchase the fertilizers from importers and sell it to members (at a discount) and to non-members.

SC5 occurs when government-owned enterprises procure fertilizers for their national fertilizer programs. They either procure it directly from overseas suppliers or via local importers using a tender system. These fertilizers are typically distributed via government parastatals or the Ministry of Agriculture extension service for free, at subsidized prices or on credit. The private sector is not involved in the distribution of fertilizers procured by the government.

1.2.2 Basic Market Conditions

Small-scale farmers produce the majority of agricultural products in SSA, but they use little or no chemical or organic fertilizers to do so. More fertilizer is applied to staple foods than to export crops; 40 % of the fertilizer consumed in SSA is used on maize, followed by other cereals (wheat, barley, teff, sorghum and millet). Fruits and vegetables and sugar cane account for 15 % of fertilizer use, and rice, tobacco, cotton and traditional tubers (cassava, yams) account for 2–3 % each (Morris et al., 2007).

Average fertilizer use in SSA is 10 kg/ha, one-tenth of the world average (100 kg/ha). Consequently, national fertilizer markets are small, with the majority of countries in SSA consuming less than 10,000 mt of nutrients per annum. These markets are characterized by too many products relative to market size. For example, the West African cotton sector has a high level of fertilizer product differentiation among different countries despite the fact that they have similar soil types. Similarly, Malawi has 20 fertilizer products in use. Many are compound fertilizers, typically NPKs with minor variations in content. Because these are low-analysis fertilizers, the nutrients are more expensive than the same nutrients found in straight high-analysis fertilizers, because it is more expensive to manufacture smaller amounts of specific types of fertilizers.

Fertilizer markets in Africa underwent liberalization to varying degrees beginning in the late 1980s/early 1990s but in many countries there is still government involvement in fertilizer importation and/or distribution to varying degrees.

- The majority of fertilizer consumed in Africa is in **granular form**. Therefore, either it is imported bagged or it is imported in bulk and bagged at the port at a flat cost per bag using equipment owned by the Port Authority and/or private bagging companies.
- In regard to handling and storage, palletization is often not used in Africa; instead **bags** are carried manually or dragged, which can result in torn bags and spillage.
- Fertilizer is **hygroscopic**, so improper storage can result in caking of fertilizer and/or its chemical composition can be affected due to exposure to heat and humidity. Therefore losses due to improper handling and storage can be quite high.
- Importation of fertilizer is **location-specific**, while in most countries its consumption is highly dispersed and located at a distance from the port. Therefore, a reliable and efficient transportation system is required to move fertilizer to farmers in Africa.
- The **road network** in Africa, particularly in rural areas, is sparse and those that do exist are poorly maintained. This results in high maintenance costs for trucks and increases travel time which increases transaction costs (in terms of time and money).

As a result, there is poor availability of fertilizers in rural areas and high prices for fertilizers that do reach the farm-gate. The seasonality of demand for fertilizers and the low purchasing power of smallholders which is typically characterized by small and frequent purchases increase transportation, storage and transaction costs.

1.2.3 Performance of African Fertilizer Markets

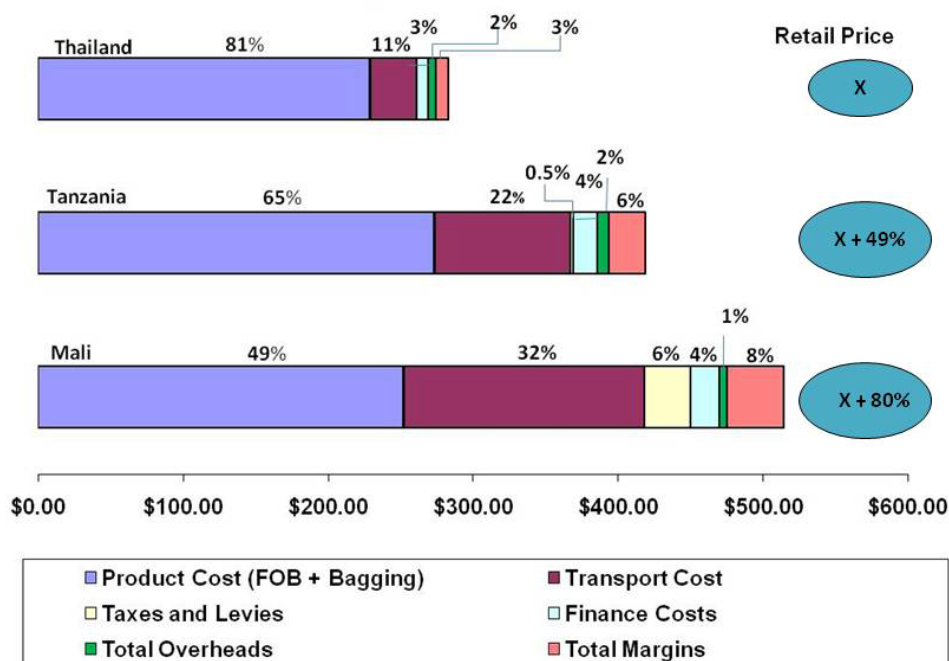
As the main actors in the fertilizer supply chain carry out functions they generate costs and also take remuneration for their services in the form of profit margins. Figure 5 illustrates fertilizer supply chains for urea in Thailand, Tanzania (a coastal country) and Mali (a landlocked country) in 2006. It illustrates the costs that are incurred as the various functions are performed along the supply chain, and illustrates their percentage contribution to the retail price.

The main cost components are the product cost (f.o.b. and bagging), taxes and levies, transportation, finance and total profit margins by traders. Comparing the value chains for the three countries provides some useful insight with respect to performance of the fertilizer market in Africa. It is evident that (in absolute terms) product costs are relatively similar in the three countries. They account for the largest component in the supply chain, representing 81 % of the retail price in Thailand (\$229/mt), 65 % of the retail price in Tanzania (\$251/mt) and 49 % in Mali (\$252/mt).

However, all other costs are much higher in Tanzania and Mali than in Thailand. Whereas these other costs account for 19 % of the retail price in Thailand, they account for 34.5 % of the retail price in Tanzania, and 53 % in Mali. Some proportions of these higher costs in Africa are legitimately incurred while performing the various marketing functions; however, some proportion of these costs are incurred as a result of the constraints imposed by the policy, regulatory and institutional environment.

Consequently, the retail price of urea in Tanzania is 50 % higher than that in Thailand and in Mali it is 80 % higher. In general, in-country costs increased the cost of fertilizer to the farmers in Africa by 60–100 % compared with 20 % in Thailand.

Figure 4: Fertilizer Cost Build Up: Thailand vs. Sub-Saharan Africa (2006)



The other cost components that are responsible for the higher in-country costs in Tanzania and Mali are transport costs, total margins and finance costs. Taxes and levies are also a major factor in Mali. Total transport costs include ocean freight and port charges, inland transport and retail transport costs. These costs account for 22 % (\$93/mt) in Tanzania and 32 % (\$165/mt) in Mali compared with 11 % (\$31/mt) in Thailand. As is to be expected, transport costs are higher in landlocked countries like Mali compared with coastal countries like Tanzania which do not have to contend with inland transport costs. In general, transport costs are \$119/mt for coastal countries in Africa and \$136/mt for landlocked countries.

The higher transport costs in Africa are due to a number of factors. First, higher ocean freights and port charges in coastal African countries (\$49/mt) compared with \$23/mt in Thailand in 2006 due to relatively smaller cargo sizes, difficulties of finding return cargo and port congestion due to poor infrastructure. For example, cargoes to Thailand are 44,000 mt compared with the majority of fertilizer trade in SSA which is in small consignments of 1,000–25,000 mt (bulk) and 5,000–10,000 mt (bagged). Second, port charges in coastal African countries (\$12/mt) are over 10 times higher than those in Thailand (\$1/mt). A contributing factor is the various “fees” charged by the port authority and customs agents at the ports and countries of destination which are undocumented. Inland transport costs are a third significant component of total transport charges. They can account for 20–40 % of the retail value of fertilizers in Africa.

Importer, wholesaler and retailer margins account for 3 % of the retail price in Thailand compared with 6 % in Tanzania and 8 % in Mali. Finance costs are the fourth-largest component in the three fertilizer supply chains depicted in Figure 7. They account for 4 % of the retail price in Tanzania and

Mali compared with 2 % in Thailand. Finance costs are comprised of the costs of obtaining letters of credit (LC), bank fees and interest rates incurred by importers, distributors and retailers. The reasons for the discrepancy are the cost of opening accounts and the cost of LC in Thailand is much lower than that in Africa; it is 0.25 % in Thailand, which is one-tenth the rate charged in Africa. Moreover, interest rates in Thailand are 5 % per annum for importers and 7 % for wholesalers and retailers compared with 25–30 % per annum in Africa.

What accounts for the higher costs and margins in African countries than in Thailand? A key reason is poor market development in Africa. Thailand has lower margins and overhead costs due to a higher level of development of retail networks in terms of quantity and quality of market actors (there is one retailer per 5,000 farmers in Thailand), fierce competition at all levels of the supply chain and the high volumes of business conducted. These attributes are absent in the fertilizer market in Africa due to a number of supply- and demand-side constraints that are inflating the magnitude of these costs above what they would be in a competitive market. That is, there is a component of each cost that can be justified by the function performed (whereby $\text{Price} = \text{Marginal Cost}$). But in African fertilizer markets the final consumer price is not equal to the marginal cost but is above it ($\text{Price} > \text{Marginal Cost}$) because there are rents or additional costs. These rents emanate due to the policy, institutional and regulatory environments in which the fertilizer market operates.

1.3 Constraints to Market Development

1.3.1 Supply-Side Constraints

A key supply-side constraint is **policy uncertainty and inconsistency**. Many African governments are unclear about their role in a liberalized fertilizer market; specifically, what they should do to support the fledgling private sector. At the same time, governments have little faith in the capability of the private sector.

The result is two extremes in the governments' approach to the fertilizer market. One extreme is a 'hands-off' approach whereby the government has withdrawn completely and left the private sector unsupported in terms of providing an enabling policy, regulatory and institutional environment. The other extreme is "government intervention," – government involvement in the importation and/or distribution of fertilizer via the reinstatement of old parastatals, typically implemented in the form of an unclear and inconsistent subsidy policy which eschews the private sector.

Where the subsidy programs do not involve the private sector, the result is market distortions due to the sale of subsidized fertilizers by the government parastatals and the displacement of commercial sales, which discourages the private sector. Even where the private sector may be requested to import on behalf of the government, the information regarding quantities and timing is not always clear or provided in a timely manner, and the terms of participation can pose risks for private importers who decide to submit bids. For example, tenders can be unrealistic as they require physical stocks to be positioned in-country prior to tendering which poses a huge risk for importers as they have no guarantee their bid will be successful.

Therefore, despite the liberalization of fertilizer markets in Africa, continued government involvement to varying degrees is creating an unequal playing field and disrupting the development of the burgeoning private sector.

Other supply-side constraints are **fertilizer regulatory systems** that are either nonexistent or ineffective where they do exist and weak market information systems. Many countries either do not have a legal and regulatory framework regarding quality, standards, measures, safety in use and business ethics vis-à-vis the importation, distribution, marketing and use of fertilizer products in the country. Where these frameworks do exist, the quality control systems are weak and implementation capacity is limited. Laboratory testing facilities are absent or outdated and the majority of countries have no inspectors or there are less than 10 inspectors for the whole country; hence inspection at the point of sale, where the risk of adulteration is highest, is limited.

Well-functioning markets require **regular and accurate information** about prices, quantities, stocks, deliveries and transaction costs. However, while in some countries information about output markets (commodity prices) is published in the daily newspaper, information collection and dissemination remain weak for input markets. The ministries of agriculture in many African countries do not have the human and/or financial resources to collect and disseminate market statistics and information. Consequently, importers, agro-dealers and farmers do not have sufficient, up-to-date information about market conditions to make intelligent decisions about where and when to buy and sell their fertilizers and other agricultural inputs.

1.3.2 Demand-Side Constraints

These consist of constrained fertilizer adoption due to the absence of stable output markets for increased production, outdated fertilizer recommendations, and weak or non-existent extension systems.

The existence of reliable and stable output markets provide the incentive for farmers to use productivity-enhancing technologies like fertilizers by providing reliable outlets for their marketable surpluses. However, while the markets for cash and export crops are well-developed in SSA, the markets for food crops are poorly developed. Consequently, at harvest farmers are often faced with low prices which reduce the incentive to use modern inputs.

Second, small-scale farmers in SSA lack knowledge about the correct and safe use of fertilizers. Very few farmers use basal fertilizers, some use basal (NPK) fertilizers for topdressing, or they may use a mixture of both, due to knowledge and economic constraints.

Moreover, even where farmers attempt to use fertilizers correctly their efforts are hampered by **outdated fertilizer recommendations**; consequently, they use fertilizer grades and quantities that are not suitable for their soils and/or crop mix. This continuous cultivation without proper and adequate use of fertilizers has resulted in severe soil infertility and degradation problems in SSA. Consequently, crop yields and profitability are much lower than what is required to achieve food security and increased incomes.

A key reason for the outdated fertilizer recommendations and poor farmer knowledge regarding correct fertilizer use in SSA is that many of the **extension and research systems** are defunct or if they exist, they are under-staffed and under-equipped.

1.4 Concluding Remarks

The establishment of well-functioning fertilizer markets in Africa will require the implementation of a set of appropriate policy, regulatory, institutional, and market development measures which will attract private sector investment and engagement by establishing a level playing field for all market actors while enabling government to ensure market access for poor farmers.

The key policy measures are: the establishment of a consistent and predictable price (subsidy) and trade (tariffs, taxes) policy; introduction of risk-sharing financial mechanisms to share risk among stakeholders thus increasing access of affordable finance for importers and agro dealers; investment in human capital development for farmers and the private sector, (marketing and technical skills); development and enforcement of fertilizer legislation and supporting the supporting regulatory framework.

On the demand side, measures should be implemented to improve the prices received by farmers and dramatically increase output market demand. Output market demand can be increased by scaling-up and replicating output markets on different fronts – out grower schemes, peri-urban agriculture, niche markets and non-traditional exports.

The prices received by farmers can be improved by promoting the development of producer associations, dissemination of market information, introduction and enforcement of grade and standards for quality produce, improved storage and agro-processing facilities and warehouse collateral to facilitate the purchase of agricultural produce at harvest time.

There is a need to educate farmers about the proper use of fertilizers and management practices. Extension systems should be revived, and redesigned and capacitated to reach the widely dispersed farming community in Africa. Farmers need to be educated about correct fertilizer use and improved agronomic practices through fertilizer trials, demonstrations and field days.

In addition, laboratories and facilities should be provided to enable soil analysis and testing to update the fertilizer recommendations and make them more appropriate to the different agro ecological zones and input and output market realities faced by farmers.

However, the key policy response which is of central interest to this training manual is the development of market information systems. Ministries of Agriculture need to establish and operate market information systems to provide information to importers, agro-dealers, farmers and policymakers about market conditions. This flow of information should be regular and timely, and the content should be accurate. In this way every market participant can have access to reliable information about prices, stocks and deliveries in various segments of national, regional and global markets, and use this information to make the right decisions.

Chapter 2. Types, Definitions, Classification, and Uses



Learning Objectives

- Participants are expected to be able to explain the difference between organic and inorganic fertilizers and why inorganic fertilizers take centre stage in analysis.
- Participants should be able to identify different types of fertilizers (Straight, Complex and Micronutrients) and their usefulness to crops.
- Participants should be able to calculate nutrients and nutrient ratios contained in a given Fertilizer product.

2.1 Defining Fertilizers and Nutrients

Fertilizers are manures, chemicals, and minerals that supply one or more essential plant **nutrients**. Application of fertilizers at the opportune time using correct quantities can ensure reasonable growth benefit to plants.

Fertilizers can be classified into two categories:

1. **Organic fertilizers:** These are obtained from living or once-living material such as animal waste (manure), crop residues (such as leaves), compost, and many other byproducts of living organisms.
2. **Inorganic fertilizers** (also called chemical or mineral fertilizers): The products are essentially originated from non-living sources through manmade processes. Most of the commercial fertilizers fall under this category.

2.1.1 Differences between Fertilizer Products and Nutrients

Fertilizer products refer to the types of fertilizers that deliver nutrients to the crops and some of them are those which can be identified through the use of HS codes. When bagged the product is given in terms of type as well as quantity. An example can be 50kg Ammonium Nitrate. A 50kg bag Ammonium Nitrate is the product in this case.

The product is silent in telling us the individual quantities of nutrients available in it. It (product) does not state the %ages of Nitrogen, Phosphorus or Potassium available. A product may contain one or more of the principal nutrients together with any other micronutrients including some non-fertilizer material. The non-fertilizer material may be sand or some such material used as a carrier for the nutrients. This would explain why the total mass of all nutrients in a given product may be different from the product mass.

2.1.2 The building blocks of plant nutrients



Organic and inorganic fertilizers can supply the **3 primary nutrients** (Nitrogen (N), Phosphorus (P) or Potash (K)), as well as **secondary nutrients** and **micronutrients**.

We can compare plant nutrients to building a wall. To build a wall you need a place (piece of land) to build in on. Then you need to build a foundation with concrete. Without a foundation the wall will crack and collapse. After the foundation you build the wall with bricks. Thereafter you cover the bricks with plaster. This is now a wall, which will not collapse.

With plant nutrition the same principle applies:

- First the plant needs the primary nutrients which form the foundation of plant nutrition: nitrogen, phosphate and potassium.
- Then the plant the secondary nutrients that can be compared to the bricks: calcium, magnesium and sulfur.
- The plant also needs the micronutrients to grow that are like the plaster. Without it the other two will not be effective: copper, iron, manganese, zinc, boron, molybdenum and chlorine

Table 2: Essential elements for fertilizers

Element	Expressed as :	Explanatory notes
Carbon	C	Elements 1 -3 exist & can be harvested as individuals
Hydrogen	H	
Oxygen	O	
Primary Nutrients		
Nitrogen	N	Exists individually. Phosphorus & Potassium are highly volatile & so exist in oxide form. They quickly react with oxygen.
Phosphorus	P ₂ O ₅	
Potassium	K ₂ O	
Secondary nutrients		
Sulphur	S	Elements 8 -9 exist as oxides
Calcium	CaO	
Magnesium	MgO	
Micronutrients		
Boron	B	These can exist as single elements
Copper	Cu	
Iron	Fe	
Zinc	Zn	

(Source: FAO, 1991)

2.1.3 Primary Fertilizer Elements & their functions in crops

The primary fertilizer elements are those that play a major role in the development of agricultural crops. They include Nitrogen (N), Phosphorus (P) and Potassium (K). Phosphorus and Potassium are both highly reactive and therefore are normally harvested in their oxidized forms, P₂O₅ and K₂O, respectively.

The different functions of these primary elements are shown in the table below.

Table 3: functions of primary elements (N P K)

Name	Benefits	Deficiency symptoms
Nitrogen (N)	Gives plant the green color, increased growth and better yields.	Leaves pale green or yellow and plants are stunted.
Phosphate (P)	Makes young plant strong, creates healthy roots, enhance overall plant quality and leads to early crop maturity.	Purplish color on older leaves, slow fruit maturity, weak seed development and plants are stunted.
Potassium (K)	Produces good seeds and fruits, contributes to general plant health, makes plant disease resistant and supports water uptake of the plant.	First sign is dark green leaves. Weak stems. Leaves later become dark brown with dying off of the tissue.

2.1.4 Secondary nutrients

The NPKs are primary nutrients whereas Calcium (Ca), Magnesium (Mg), Sodium (Na) and Sulphur (S) are secondary nutrients whose function is for the nutrition of plants.

Table 4: Secondary nutrients in some commonly used fertilizers

Fertilizer and amendments	% Sulphur as S	% Calcium as CaO	% Magnesium as MgO
Fertilizer and amendments			
Ammonium sulphate	24		
Calcium ammonium nitrate		10 - 20	0 - 7.5
Single superphosphate	12	25 - 30	0.5
Triple superphosphate	1	17 - 20	0.5
Potassium sulphate	18	-	-
Potassium chloride with magnesium	4	-	6
Potassium magnesium sulphate	18	-	8 - 11
Epsom salt	13	-	16
Diammonium phosphate	up to 3	-	-
Ammonium phosphate sulphate	10 - 14	-	-
Dolomitic limestone	-	29	5 - 20
Kieserite	20 - 22	-	26 - 27
Quick lime/burnt lime/unslaked lime	-	85	-
Gypsum (calcium sulphate)	15 - 18	23	-
Micronutrient fertilizers			
Zinc sulphate	18	-	-
Manganese sulphate	15	-	-
Copper sulphate	13	-	-
Iron sulphate	12	-	-

2.1.5 Micronutrients

Micronutrients include elements such as Boron (B), Copper (Cu), Iron (Fe), Zinc (Zn), Manganese (Mn), Molybdenum (Mo) and Chlorine (Cl). Which are vital for health in human beings and are taken in minute quantities by plants.

The micronutrient chlorine, for example, is delivered to plants through application of Potassium Chloride (KCl) also known as Muriate of Potash.

HOW TO REMEMBER THE PLANT NUTRIENTS

CONCRETE FOUNDATION = Nitrogen, Phosphate and Potassium (Primary Nutrients)

BRICKS = Calcium, Magnesium and Sulfur (Secondary Nutrients)

PLASTER = Iron, Copper, Boron, Manganese, Zinc, Chlorine and Molybdenum (Micronutrients)

2.1.6 Converting oxides

For the elements that exist as oxides it is important to compute the nutrient content of the elements; P, K, Ca and Mg. The conversion factors are given below. An example of how these contents have been found is also given in the case of potassium.

$P_2O_5 * 0.4364 = P$	$P * 2.2919 = P_2O_5$
$K_2O * 0.8302 = K$	$K * 1.2046 = K_2O$
$CaO * 0.7147 = Ca$	$Ca * 1.3992 = CaO$
$MgO * 0.6030 = Mg$	$Mg * 1.6582 = MgO$

2.2 Organic Fertilizers

Organic manures are some form of fertilizers that are derived from crop residues, weed biomass and animal waste including sewage wastes. There are many ways suggested to improve the quality of organic fertilizer and these include planting legumes in order to improve nitrogen content in animal dung, pit storage of manure to prevent leaching and drying of nutrients as well as anaerobic treatment of manure.

Table 5: nutrient contents of fertilizer of animal origin

Material	Nitrogen (N%)	Phosphate (P ₂ O ₅ %)	Potash (K ₂ O %)
Dried blood	10.0 - 12.0	1.0-1.5	0.6 - 0.8
Fish manure	4.0 - 10.0	3.0-9.0	0.3 - 1.5
Bird guano	7.0 - 8.0	11.0-14.0	2.0 - 3.0
Hoof and horn meal	14	1	
Bone meal (raw)	2.0 - 4.0	22.0 - 24.0	
Bone meal (steamed)		22.0 - 30.0	

Organic fertilizers have been and is still are being used in many rural SSA countries as cost of production of them is low. However, the problem with organic manure is that of low content of plant nutrients and being bulky making transportation difficult.

Since the nutrient content of organic fertilizers is low, they cannot be relied upon exclusively to raise agricultural yields substantively. It is imperative to use mineral or inorganic fertilizers to provide plants with sufficient quantities of nutrients that they can access and use efficiently. However, mineral fertilizers must be used in conjunction with organic fertilizers since the later have other critical characteristics which are advantageous to plant growth.

Table 6: Summary of the different organic fertilizers

Name	Remarks
Manure	Complete fertilizer (N+P+K) – nutrients may vary depending on what the animals eating. It is rather used as soil conditioner than fertilizer.
Sewer sludge	Available in a dry granular form from sewage treatment plants. Not advisable for use on edible crops if not composted.
Fish meal	High in nitrogen and micronutrients. Strong odor.
Seaweed emulsion	Excellent source of nutrients included micronutrients but tends to be expensive.
Bone meal & blood meal	Derived from animal abattoir leftovers. Bone meal is high in phosphate, calcium and a bit of nitrogen. Blood meal provides nitrogen.
Cotton or oil seed meal	Source of nitrogen.

2.3 Inorganic Fertilizers



Inorganic fertilizers are those that are manufactured with **pre-determined or declarable quantities of Nitrogen, Phosphorus and Potassium (NPK) nutrients**. These can be straight fertilizers with only one of the nutrients or compounds with a mixture of two or more nutrients.

2.3.1 Forms of inorganic Fertilizers

As a way of introduction it is imperative for us to be aware of the different forms of mineral fertilizers for the purposes of matching crop product and the fertilizer required.

Table 7: Different forms of inorganic fertilizers

Fertilizer class	Description	Example(s)	Notes
Straight Fertilizers	These fertilizers have known content of one of the three Primary nutrients, N, P & K	(NH ₄) ₂ SO ₄ Ammonium Sulphate	Used in alkaline soils to raise pH of soil & supplying N
		(NH ₄)NO ₃ Ammonium Nitrate	Supplies N to plants & promotes vegetative growth. Does not interfere with pH of soil
		Ca(NO ₃) ₂ Calcium Nitrate	Considered the best type of Fertilizer as absorption of Ca & N is good. Best for Greenhouse use
Compound Fertilizers	The fertilizers have more than one of the three primary nutrients, (NP, NK, PK & NPK)	(NH ₄) ₂ HPO ₄ - Diammonium phosphate (DAP)	Increases soil pH (Prolonged use causes pH value to increase beyond optimal.
		NH ₄ H ₂ PO ₄ - Monoammonium phosphate (MAP)	Supplies soil with nitrogen & phosphorus
Ca, Mg, Na, S Fertilizers	They are not NPKs		Just for nutrition of plants
Inorganic soil conditioners	Liming materials. Contains one or two elements Ca & Mg	CaSO ₄ .2H ₂ O – calcium sulphate dehydrate (Gypsum)	Used to maintain or raise soil pH value.

Source: (Adapted from FAO Manual on Fertilizer Statistics, 1991)

2.3.2 Commonly used fertilizers

Table 8: composition of some important fertilizers

COMMON NAMES		N	P ₂ O ₅	K ₂ O	S	MgO
(Nutrients as % of product)						
Nitrogen Fertilizers						
Ammonia		82	0	0	0	0
Ammonium sulphate	AS	21	0	0	23	0
Ammonium nitrate	AN	33-34.5	0	0	0	0
Calcium ammonium nitrate	CAN	20.4-27	0	0	0	0
Urea		45-46	0	0	0	0
Phosphate Fertilizers						
Single superphosphate	SSP	0	16-20	0	12	0
Triple superphosphate	TSP	0	46	0	0	
Diammonium phosphate	DAP	18	46	0	0	
Monoammonium phosphate	MAP	11	52	0	0	
Ground rock phosphate		0	20-40	0	0	
Potash Fertilizers						
Muriate of potash	MOP	0	0	60	0	0
Sulphate of potash	SOP	0	0	50	18	0
Sulphate of potash magnesia		0	0	22-30	17-22	10-11
Magnesium Fertilizers						
Kieserite		0	0	0	20-22	25-27
Epsom salts		0	0	0	12-13	15-16
Complex Fertilizers						
NPK fertilizers	NPK	5-25	5-25	5-25	*	*
NP fertilizers	NP	15-25	15-25	0	*	0
NK fertilizers	NK	13-25	0	15-46	*	0
PK fertilizers	PK	0	7-30	10-30	*	*

* Some with S and/ or Mg and/or micronutrients

The plant nutrients phosphate and potassium may be expressed as their elemental forms P and K or as their oxide forms P₂O₅ and K₂O. Throughout this manual we use the oxide form. Nitrogen is expressed as N.

2.4 Classification of fertilizers (HS codes)

What is the Harmonized System?

HS stands for Harmonized Commodity Description and Coding System. The HS is the international standard for reporting goods to customs and other government agencies. It is a numeric language that is used by more than 180 countries worldwide, and almost 100% of international trade. The HS was created and is administered by the Brussels-based World Customs Organization (WCO).

What are HS Codes?

HS codes are the language of international trade. Every commodity that enters crosses an international border must be declared to customs by means of a specific Harmonized System code. Among other uses, these numerical codes are used to calculate and assess duties and taxes, determine import and export admissibility, and conduct risk assessment and targeting. The first 6 digits of the HS are used universally. *Each country may then add to the original 6 to suit its own tariff and statistical needs, creating 8, 10, and sometimes 12 digit national codes.*

Country-specific modifications of fertilizers

Slight modifications of imported fertilizers by countries do occur resulting in country-specific names for such products. An example could be Compound D fertilizer in Malawi. This fertilizer is an NPK 8-18-15 +6s +0.1B type of fertilizer with additions of Sulphur and Boron. Compound D fertilizers are a group of NPKs with slight variations and mostly used for cereals. The analogue of this can be found in various petroleum products in which unleaded petrol is slightly different from one garage to the other in terms of content.

Common HS codes for fertilizers

Table 9: Common HS codes for fertilizers

Heading	HS Code	Fertilizers
31.01		Animal/Vegetable Fertilizer, mixed/not/chemically treated
	3101.00	Animal or Vegetable Fertilizers, Fertilizers Produced By the Mixing
31.02		Mineral or chemical fertilizers, nitrogenous
	3102.10	Urea (Whether or Not in Aqueous Solution)
	3102.21	Ammonium Sulphate
	3102.29	Double Salts and Mixtures of Ammonium Sulphate or Ammonium Nitrate
	3102.30	Ammonium Nitrate
	3102.40	Mixtures of Ammonium Nitrate
	3102.50	Sodium Nitrate
	3102.60	Double Salts and Mixtures of Calcium Nitrate and Ammonium Nitrate
	3102.70	Calcium Cyanamide
	3102.80	Mixtures of Urea and Ammonium Nitrate in Aqueous or Ammoniacal Solution
	3102.90	Other Nitrogenous Fertilizers
31.03		Mineral or chemical fertilizers, phosphatic
	3103.10	Superphosphates
	3103.20	Basic Slag
	3103.90	Other Phosphatic Fertilizers
31.04		Mineral or chemical fertilizers, potassic.
	3104.10	Carnallite, Sylvite and Other Crude Natural Potassium Salts
	3104.20	Potassium Chloride
	3104.30	Potassium Sulphate
	3104.90	Other Potassic Fertilizers
31.05		Mineral or chemical fertilizers containing two or three of the fertilizing elements nitrogen, phosphorus and potassium
	3105.10	Fertilizers in Tablets or in Packages of a Gross Weight Not Exceeding 10kg
	3105.20	Mineral or Chemical Fertilizers Containing Nitrogen, Phosphorus, Potassium
	3105.30	Diammonium Hydrogen orthophosphate (Diammonium Phosphate)
	3105.40	Ammonium Dihydrogen orthophosphate, Mixtures Thereof With Diammonium Hydrogen orthophosphate
	3105.51	Fertilizers Containing Nitrates and Phosphates
	3105.59	Other Mineral or Chemical Fertilizers Containing Nitrogen, Phosphates
	3105.60	Mineral or Chemical Fertilizers Containing Phosphorus and Potassium
	3105.90	other Fertilizers
28.14		Ammonia, anhydrous
25.10		Phosphate rock

Source: http://www.wcoomd.org/home_hsnomenclaturetable2012.htm

2.5 Converting fertilizer products in nutrients

2.5.1 Computation of Nutrient content for Fertilizers in General

It is of utmost importance to know the content of each nutrient in a given fertilizer. This is important for the purpose of applying the right amount in order to optimize on harvesting. Incorrect use may result in undesirable outcomes that include over-fertilization or underuse of the product.

2.5.1.1 The nutrient content analogue in the real world

The computation for nutrient content for Fertilizers is about calculating the content of N, P and K available in a given fertilizer which may be a straight or compound fertilizer.

By way of formula, if you are given X kg of a Fertilizer and that $Z\%$ of it is Nitrogen then the nitrogen content is computed as follows:

$$\text{Amount of Nitrogen in the fertilizer (kilograms)} = \frac{Z}{100} * X \text{ (kg)}$$

*Example 1: if one has 200 metric tons (200, 000kg) of Ammonium nitrate and that it contains 21% of Nitrogen, then the Nitrogen nutrient in the fertilizer is $\frac{21}{100} * 200,000 = 42,000\text{kg}$.*

Example 2: f a 50kg bag of Ammonium Nitrate contains 21% Nitrogen,

$$\text{then Nitrogen content} = \frac{21}{100} * 50 = 10.5\text{kg}$$

2.5.2 Fertilizer Grade (Analysis)

The **grade** of a fertilizer refers to the **guaranteed minimum %age(s) of the nutrients** it has and is expressed for N, P, and K (and if more when secondary micronutrients are known).

- Nitrogen concentration in Ammonium Nitrate is 21% (21-0-0) and therefore 100kg of Ammonium Nitrate contains 21kg of Nitrogen.
- Similarly, Diammonium phosphate (DAP) contains 18% N, 46% P_2O_5 and so the grade for Diammonium phosphate is 18-46-0.

Generally a compound that contains $\alpha\%$ N, $\beta\%$ P and $\mu\%$ K is graded α - β - μ .

Some common examples are NPK 17 17 17, NPK 15 15 15, NPK 20 20 0.

2.5.3 Conversion table from products to nutrients

Table 10: conversion table for common fertilizers

HS Code	FERTILIZERS	Short name	N	P2O5	K2O
3102	Nitrogen Fertilizers				
28.14	Anhydrous Ammonia	(liquid)	82		
3102.10	Urea		46		
3102.21	Ammonium Sulphate	AS	21		
3102.30	Ammonium Nitrate	AN	33.5		
3102.50	Sodium Nitrate		16		
3102.40	Calcium Ammonium Nitrate	CAN	26		
3102.80	Urea Ammonium Nitrate	UAN	32		
3103	Phosphate Fertilizers				
25.10	Rock Phosphate			20-40	
3103.1	Basic Slag			14	
3103.10	Single Superphosphate	SSP		18	
3103.1090	Triple Superphosphate	TSP		46	
3104	Potash Fertilizers				
3104.2	Potassium Chloride (or muriate of potash)	MOP			60
3104.3	Potassium Sulphate	SOP			50
3105	Complex Fertilizers				
3105.70	NK fertilizers	NK	13-25	0	15-46
3105.10/20	NPK fertilizers	NPK	5-25	5-25	5-25
2834.21	Potassium Nitrate	NK	13		47
3105.30	Di ammonium Phosphate	DAP	18	46	
3105.40	Mono ammonium Phosphate	MAP	11	52	
3105.51	NP fertilizers	NP	15-25	15-25	0
3105.60	PK fertilizers	PK	0	7-30	10-30



Fertilizer Definitions - Highlights

- Fertilizers are manures, chemicals or minerals that deliver nutrients to plants.
- Fertilizers can be organic or inorganic in nature.
- Inorganic fertilizers have declarable quantities.
- Fertilizers contain one or more of the primary nutrients N (nitrogen), P (phosphate) or K (potash) and in some cases secondary nutrients and micronutrients.
- Straight fertilizers contain only one primary nutrient; compound fertilizers are a mixture of 2 or more primary nutrients
- It is important to know and calculate the nutrient concentration in a given fertilizer. Fertilizer products are converted into nutrient contents for statistical uses.
- The HS codes are used to identify fertilizers and are grouped under class 31.
- HS codes are very useful when receiving products at the customs offices at the port of entry.



Quiz

1. What is the difference between organic and inorganic fertilizers
2. Give one example of primary fertilizer, secondary fertilizer and micronutrient
3. Given a 50kg bag of an NPK fertilizer with the following specifications NPK 23-10-5
 - (i) Calculate fertilizer nutrient quantities
 - (ii) Give the HS code for the fertilizer mentioned above.

Chapter 3. Fertilizer “hard data”: Production, Trade, Consumption, and Price Variables



Chapter Learning Objectives

- Participants should be able to explain the important Agricultural Variables such as Fertilizer production, Fertilizer Imports, Fertilizer Exports and Fertilizer Consumption
- Participants should be able to collate and compile statistics using various templates used by FAO, IFA and CountrySTAT
 - Production
 - Imports and exports
 - Consumption
 - Prices

The focus of this chapter is on variables for which we collect purely quantitative data – so called ‘Hard’ data. ‘Hard’ data is quantitative data collected for variables such as production, consumption, prices, trade volumes and subsidies. *It is largely inspired by the FAO manual on fertilizer statistics (1991).*

3.1 Hard Data Variables

The ‘Hard’ data Fertilizer variables consist of the following:

Fertilizer Production

Fertilizer Production is the actual making or manufacturing of fertilizer from raw materials. In some cases the so-called raw materials may be straight fertilizers being used to make compound fertilizers.

Fertilizer Trade (Imports & Exports)

Importation of fertilizer is the deliveries of these products from abroad in order to ensure adequate supply of fertilizers for the farmers. Exports of fertilizer take place when we either export part of the fertilizer we produce or re-exportation of fertilizer products in transit. Exports of fertilizer products does not always mean we have excess of the product but that we may be fulfilling some bilateral trade arrangements we may be having with our trading partners.

Fertilizer Consumption (Fertilizer utilization)

Fertilizer consumption refers to the actual use of the products on crops. This variable is difficult to get and so we use its proxy which is the apparent consumption obtained through the use of the Fertilizer Resource Utilization Account (FRUA)

Non-fertilizer use

Non-Fertilizer use is use of fertilizer derivatives to make pesticides or some other unwelcome uses such as manufacturing bombs.

Fertilizer Prices

Fertilizer prices are crucial as they have a huge influence on the demand and hence consumption of fertilizer.

All the above variables are connected to each other through the **Fertilizer Resource Utilization Account (FRUA)** given below.

3.2 Fertilizer Resource Utilization Account (FRUA)

The FRUA simply shows a summary of the important agricultural variables like Production, Trade (Imports & Exports), Non-Fertilizer use and Consumption for the NPKs. The FRUA ensures that all the values of the data are accounted for by way of an '*accounting*' equation.

As given in the FAOSTAT (2011) document:



$$\text{Production (P) + Imports (M) – Exports (X) – Non-Fertilizer use (NF) = Consumption (C)}$$

$$P + M - X - NF = C$$

The Non Fertilizer Use (NF) figure may be zero if all fertilizers available are for crop production. The accounting equation means that we can get either Production or Consumption as residuals when supplied with other data. The residual production or consumption figures are known respectively as Apparent Production and Apparent Consumption.

3.3 Importance of Hard Data

Having such hard data helps all the fertilizer actors to:

- Financially plan ahead of the season
- Timeouts ordering of fertilizers by traders
- Forecast future needs of inputs
- Intelligent purchasing by farmers
- Policy making
- Implementation of subsidy programs



Chapter Highlights

- The Fertilizer Resource Utilization Account (FRUA) can be used to find apparent Consumption (C) given production (P), trade volumes [exports (X) & Imports (M)] and non-fertilizer use amount on Fertilizer (NF).
- $P + M - X - NF = C$

3.4 Fertilizer production

3.4.1 Production of fertilizers: avoiding double counting

Clear definitions of the kind of producer help to avoid problems of the double counting of nutrients.

Primary producers are those who use feedstocks, raw materials and/or intermediates to produce finished fertilizers.

Feedstocks is a term that refers to the main raw materials used in the manufacture of a product. In the case of fertilizers the main raw materials are Nitrogen (N), Potassium Oxide (K_2O) and Phosphorus Pentoxide (P_2O_5).

Secondary producers are those who use finished fertilizer materials to produce other fertilizers.

Only the production of the **primary producers** should be included in the production data, otherwise the same material will be counted twice (double counted).

Straight fertilizers and ammonium phosphate, domestically produced and imported, are often used to manufacture other fertilizers. It is important that the production statistics should not include these products twice, both as straights and then again as the compounds from which they are produced.

Urea, for example, is sometimes used to produce or enrich grades of compound fertilizers. The urea may be produced by the manufacturer, imported or procured from other domestic urea manufacturers. A manufacturer tends to measure his total production of N including the contribution from urea. The quantity of urea (from either own production or an external supply) so used should not, in fact, be included when reporting the total N produced by the manufacturer. Otherwise it is double counted.

Gross production is defined as the production of all fertilizers produced in a country from feedstocks, intermediates, raw materials or from other finished fertilizers, whether these are home produced or imported.



Net production comprises production from feedstocks, raw materials or intermediates, whether produced locally or imported, but excluding products manufactured from other fertilizers. **It is the net production figure which should be used for statistical reporting purposes.**

It is recommended that the production figures of the straight fertilizers and the ammonium phosphates should relate to their total production, whether they are sold as such or are further processed to make compounds and, of course, that the production statistics for compound fertilizers should not include these quantities.

Compound fertilizers produced from imported straight fertilizers or ammonium phosphate, should also not be included in the production figures.

3.4.2 Other production data

Production Capacities

Information on the production capacities of domestic plants should be assembled at least once a year by the central agency. Proposals for new production capacities and for the expansion of existing capacities should also be recorded.

Secondary manufacturers

Although the production and deliveries of the secondary producers should not be included in the national statistics, because of double-counting, it may be desirable to assemble separate information on the production and deliveries of the secondary producers if the mixtures or blends they manufacture play an important role in the national fertilizer market.

The production of raw materials and intermediates

It will normally be necessary to record also the production of the major raw materials and intermediates, i.e. ammonia, phosphate rock (by grade), phosphoric acid, and sulfur.

Ammonia and phosphoric acid purchased from outside sources should evidently be excluded from the national production figures of these materials.

In practice, it is difficult to differentiate between quantities of raw materials and intermediates intended for fertilizer use, and quantities intended for non-fertilizer use.

3.4.3 Production tables from the primary manufacturers

Normally production and related statistics are assembled by a governmental coordinating agency, which:

- collects data from the primary manufacturers on behalf of the Government,
- keeps a cumulative record,
- issues a consolidated report, recording the situation at a national level.

A specimen reporting form for manufacturers, monthly production, delivery and stock statistics is given in the table below. In most countries, the deliveries will need to be broken down by region. This information can be given in a separate table.

Table 11: FAO template for production data

MANUFACTURER MONTHLY REPORT
PRODUCTION, STOCKS AND DOMESTIC DELIVERIES

Month.....

To be submitted by.....

Name of the manufacturer.....

Location.....

Metric tons

PRODUCT DESCRIPTION	% NUTRIENT	OPENING STOCK	QUANTITY PRODUCES (*)	QUANTITY DELIVERED	CLOSING STOCK
Total nutrients					
N					
P ₂ O ₅					
K ₂ O					

(*) Excluding quantities manufactured from fertilizers purchased from outside sources

Table 12: Fertilizer production for selected countries (2005-2009, in mt of nutrients)

		2005	2006	2007	2008	2009
Algeria	Nitrogen N	113,180	27,440	137,780	11,040	6,600
	Phosphate P ₂ O ₅	20,430	24,310	42,610	11,580	5,460
	Potash K ₂ O					
	NPK	133,610	51,750	180,390	22,620	12,060
Egypt	Nitrogen N	1,568,080	1,833,710	2,447,910	2,618,560	2,723,590
	Phosphate P ₂ O ₅	310,080			312,051	265,000
	Potash K ₂ O					
	NPK	1,878,160	1,833,710	2,447,910	2,930,611	2,988,590
Morocco	Nitrogen N	264,100	298,940	250,215	293,601	283,500
	Phosphate P ₂ O ₅	1,042,130	1,154,570	1,140,539	791,319	1,054,180
	Potash K ₂ O					
	NPK	1,306,230	1,453,510	1,390,754	1,084,920	1,337,680
Nigeria	Nitrogen N		20,821	12,505	12,500	
	Phosphate P ₂ O ₅		12,540	6,553	6,500	
	Potash K ₂ O	3,066	14,314	6,803	6,800	
	NPK	3,066	47,675	25,861	25,800	
South Africa	Nitrogen N	281,482	166,900	154,149	118,081	220,466
	Phosphate P ₂ O ₅	313,462	130,500	54,306	156,549	199,155
	Potash K ₂ O					
	NPK	594,944	297,400	208,455	274,630	419,621
Tunisia	Nitrogen N	251,730	242,430	228,570	210,752	250,326
	Phosphate P ₂ O ₅	915,980	884,120	840,850	835,160	855,491
	Potash K ₂ O					
	NPK	1,167,710	1,126,550	1,069,420	1,045,912	1,105,817
Zimbabwe	Nitrogen N	41,485	37,129	37,200	37,200	37,200
	Phosphate P ₂ O ₅	17,106	25,910	26,000	26,000	26,000
	Potash K ₂ O					
	NPK	58,591	63,039	63,200	63,200	63,200
Total		5,142,311	4,873,634	5,385,990	5,447,693	5,926,968

Source: www.africafertilizer.org, derived from FAO

Table 13: fertilizer production template (FAO)

 FOOD AND AGRICULTURE ORGANIZATION OF THE UNITED NATIONS - STATISTICS DIVISION																					
AGRICULTURAL RESOURCES - FERTILIZER PRODUCTION																					
- The time reference period is calendar year. - Please report data in Metric Tons. If a different unit is used please indicate it, as well as any other relevant note specific to a given data, in the NOTES column. - For item descriptions and for other important instructions please refer to the "Explanatory notes" and "Instructions" sheets.																					
HS CODE	COMMODITY	A			B				C = (A/100)*B												NOTES e.g. different year used, different unit used, preliminary data reported, etc.
		Plant nutrient content %			FERTILIZER PRODUCTION (metric tons of product)				FERTILIZER PRODUCTION (metric tons of nutrient content)												
		N	P ₂ O ₅	K ₂ O	2006	2007	2008	2009	2006			2007			2008			2009			
STRAIGHT NITROGENOUS FERTILIZERS																					
2814	Ammonia, anhydrous (82%N will be used if no information is provided)																				
310230	Ammonium nitrate (33%N will be used if no information is provided)																				
310221	Ammonium sulphate (21%N will be used if no information is provided)																				
310240	Calcium ammonium nitrate (26%N will be used if no information is provided)																				
310210	Urea (46%N will be used if no information is provided)																				
310280	Urea and ammonium nitrate solutions (32%N will be used if no information is provided)																				
STRAIGHT PHOSPHATIC FERTILIZERS																					
3.1E+07	Superphosphate above 35% (46%P ₂ O ₅ will be used if no information is provided)																				
3.1E+07	Superphosphate, other (18%P ₂ O ₅ will be used if no information is provided)																				
2510	Phosphate rock (30%P ₂ O ₅ will be used if no information is provided)																				
STRAIGHT POTASSIC FERTILIZERS																					
310420	Potassium chloride (Muriate of potash) (60%K ₂ O will be used if no information is provided)																				
310430	Potassium sulphate (50%K ₂ O will be used if no information is provided)																				
COMPOUND FERTILIZERS																					
310530	Diammonium phosphate (DAP) (18%N and 46% P ₂ O ₅ will be used if no information is provided)																				
310540	Monoammonium phosphate (MAP) (11%N and 52% P ₂ O ₅ will be used if no information is provided)																				
310551 and 310559	Other NP compounds (20%N and 25% P ₂ O ₅ will be used if no information is provided)																				
310510 and 310520	NPK complex (15%N, 15% P ₂ O ₅ and 15% K ₂ O will be used if no information is provided)																				
310510 and 310520	NPK blends (15%N, 15% P ₂ O ₅ and 15% K ₂ O will be used if no information is provided)																				
310560	PK compounds (20% P ₂ O ₅ and 25% K ₂ O will be used if no information is provided)																				
283421	Potassium nitrate (13%N and 45% K ₂ O will be used if no information is provided)																				
OTHER FERTILIZERS (specify below)																					

- Reported by MT of products and nutrients, by HS class / NPK are aggregated, but can be disaggregated in this form

Table 14: Intermediary table (FAO)

FOOD AND AGRICULTURE ORGANIZATION OF THE UNITED NATIONS - STATISTICS DIVISION																			
AGRICULTURAL RESOURCES - FERTILIZER DOMESTIC AVAILABILITY																			
</																			

- Reported by MT of product, by HS class / NPK are aggregated, but can be disaggregated in this form



Fertilizer Production - Highlights

- Primary producers are those who use raw materials and/or intermediates to produce finished fertilizers. Secondary producers are those who use finished fertilizer materials to produce other fertilizers (mixtures and blends).
- It is the net production figure from primary producers which should be used for statistical reporting purposes.
- Report on production and deliveries of secondary producers for NPK and other mixtures and blends when they play an important role in the national fertilizer market.
- Production capacities of domestic plants should be assembled by product at least once a year.
- Production data shall be computed in both fertilizer products and nutrients



Quiz

In a country, 2 fertilizer plants have produced last year a total of 500,000 mt of urea and 250,000 of AS. One has also blended 50,000 mt of NPK 15-15-15.

(i) Which figures should be reported to FAO?

(ii) Fill the corresponding FAO production table (table 13)

Fertilizer import and export

3.4.4 Imports and exports

Trade statistics comprise data on imports and exports, which represent transfers of materials from one country to another. Surplus countries export to deficit countries.

Trade statistics are generally well documented in both importing and exporting countries. They normally originate from the customs service or from a state trading organization established by the government to export or import different commodities, including fertilizers. Ministries of Industry or of Agriculture are sometimes assigned the task of keeping records of imports and exports. The situation varies from country to country. Specimen formats are given in Table 16 and Table 17 .

For international trade statistics, international codes are attributed to the products (for example, the "Standard International Trade Codes" or the "Brussels Convention Codes"). These two sets of codes are increasingly being replaced by the "harmonized commodity codes", a list of which is given in Table 10: Common HS codes for fertilizers.

The importing country should tabulate the import statistics each month, by country of origin, by product, by port, whether in bagged or bulk form and with value of imports, by product. International information on fertilizer trade is available from consultants and specialized publications.

The import value is normally on a CIF basis i.e. cost, freight and Insurance. This represents the FOB (free on board) cost plus the sea freight plus insurance. Some contracts are established on a FOB

basis, with the importing countries making their own arrangements for shipping the material from the exporting countries.

3.4.5 Re-export

Substantial quantities of fertilizers are shipped to countries which are not their final destination. This may be because the port is a convenient (perhaps the only) point of entry, or because the material is imported in bulk and bagged before being re-exported, or for political reasons etc. Sometimes fertilizers are received as part of an exchange agreement and part of the quantity is re-exported.

Re-exported material is in principle, should not be included in the import nor in the export figures of the transit country. The situation is less serious if the material is recorded in both the import and in the export statistics of the transit country, since the figures cancel out in the net trade balance and the informed user of the statistics will not be misled. But sometimes this is not the case.

Re-exported materials are fertilizers that have been received from abroad and then re-packaged. After re-packaging into smaller quantities then the fertilizer can be exported to other countries. Goods in transit are not re-packaged but are forwarded unopened to the final destinations.

Wherever possible, the recorded destination of the material should be its final destination; otherwise the trade statistics give a false picture of the movements.

3.4.6 Import and export of fertilizer raw materials

Many countries both import and export fertilizers. On the other hand, trade in fertilizing raw materials and intermediates tends to be one way. Therefore, data collection for these materials is more straightforward. For raw materials, the terms "deficit" (production less than requirement) or "surplus" (production more than requirement) are not strictly valid.

A raw material producing country normally will not have adequate fertilizer manufacturing capacity to utilize all the raw materials produced and all complementary raw materials necessary to manufacture fertilizer may not be available in the country.

Data on the import or export of phosphate rock should be compiled by grade, expressed as % BPL (*),¹ and by country and may be monthly, quarterly or annual. Countries which have inadequate indigenous resources and which import the balance to supplement the shortfall are required to maintain both production (mined) and import data

1 (*) = "Bone Phosphate of lime": 1% of BPL = 0.458% P₂O₅.

3.4.7 Trade Tables

3.4.8 Collection tables

Table 15: import / export tables (FAO)

TRADE STATISTICS

DESTINATION OF EXPORTS (domestically produced)

Month.....

Product.....

Code.....

Metric tons

COUNTRY OF DESTINATION	QUANTITY (*)	NUTRIENT CONTENT % N-P ₂ O ₅ -K ₂ O	N	P ₂ O ₅	K ₂ O	VALUE (FOB)
Country (1)						
Country (2)						
Country (3)						
Country (4)						
Country (5)						

(*) Excluding imported material

ORIGIN OF IMPORTS

Month.....

Product.....

Code.....

Metric tons

COUNTRY OF DESTINATION	QUANTITY (*)	NUTRIENT CONTENT % N-P ₂ O ₅ -K ₂ O	N	P ₂ O ₅	K ₂ O	VALUE (CFR)
Country (1)						
Country (2)						
Country (3)						
Country (4)						
Country (5)						

(*) Excluding material intended for re-export.

The above tables serve as a guide but individual countries have their own templates they can use before filling the FAO template. Below are imports records from the Kenyan Customs as well as from the CountryStat in Ghana.

Intermediary tables (FAO questionnaire – table 18)

FAO captures import and export data in the “fertilizer domestic availability” table.

Table 16: Detailed import records (KBC / CountrySTAT Kenya, May 2009)

HS CODE	PRODUCT DESCRIPTION	QUANTITY	CIF VALUE	ORIGIN	IMPEXP PIN	IMPORTER NAME
31021000	UREA	1	10,154	UNITED KINGDOM	P051094812Q	WELLCOME TRUST RESEARCH LABORATORIES
31021000	UREA 46 PERCENT IN BULK	8,800,000	232,751,600	RUSSIAN FEDERATION	P051104351N	YARA EAST AFRICA LIMITED
31022900	INCA AGRICULTURAL FERTILIZER PACKED IN 10 LITRE CONTAINERS	940	445,748	UNITED KINGDOM	P051132393W	LACHLAN KENYA LIMITED
31022900	NUTRI TECH TRIPLE TEN AGRICULTURAL FERTILIZER	2,660	1,109,233	SOUTH AFRICA	P051132393W	LACHLAN KENYA LIMITED
31022900	VITAZyme AGRICULTURAL FERTILIZER	1,734	1,400,466	UNITED STATES OF AMERICA	P051132393W	LACHLAN KENYA LIMITED
31023000	AMMONIUM NITRATE	10	70,592	NETHERLANDS	P000595412P	OSERIAN DEVELOPMENT CO LTD
31023000	AMMONIUM NITRATE POROUS PRILLS	20,000	776,419	IRAN	P051151500Y	NITRO CHEMICALS LIMITED
31026000	CALCIUM NITRATE	50,000	1,331,004	CHINA	P051113464S	LIKI RIVER FARM LIMITED
31029000	CALCIUM AMMONIUM NITRATE CAN 26 PERCENT N FERTILIZER IN BULK	15,750,000	292,846,688	ITALY	P051104351N	YARA EAST AFRICA LIMITED
31029000	4,400 MTS CALCIUM AMMONIUM NITRATE CAN 26.9 CAO FERTILIZER IN BULK	4,400,000	100,818,008	SPAIN	P000630680S	SUPPLIES SERVICES LTD
31039000	DIAMMONIUM PHOSPHATE (18-46-0) IN BULK	1,000,000	45,099,148	UNITED STATES OF AMERICA	P000596838I	EXPORT TRADING CO LTD
31039000	DIAMMONIUM PHOSPHATE (18-46-0) IN BULK	5,000,000	225,495,744	UNITED STATES OF AMERICA	P000596838I	EXPORT TRADING CO LTD
31049000	HUMATE POTASSIUM	10,800	1,440,035	CHINA	P051176196L	HANGZHOU AGROCHEMICAL IND EA LIMITED
31052000	AGROLEAF FERTILIZER	20,790	4,963,690	NETHERLANDS	P051130567S	THE SCOTTS COMPANY KENYA LIMITED
31052000	NPK COMPOUND FERTILIZER WITH 2 PERCENT EMPTY BAGS	28,500,000	781,167,168	ROMANIA	P000593739A	KENYA TEA DEVELOPMENT AGENCY LIMITED
31052000	590 CARTONS 590 BAGS OF AGROLEAF POWER CALCIUM 11.05.19.2.51	15,300	3,365,966	NETHERLANDS	P051130567S	THE SCOTTS COMPANY KENYA LIMITED
31052000	2640 BAGS OF AGROLEAF POWER 40-40-40 TE FERTILISERS (FERTILISER)	39,600	7,142,149	NETHERLANDS	P051130567S	THE SCOTTS COMPANY KENYA LIMITED
31052000	AGROLEAF POWER FERTILISER	14,256	2,730,737	NETHERLANDS	P051130567S	THE SCOTTS COMPANY KENYA LIMITED
31052000	AGROLEAF POWER FERTILISER	14,160	3,143,217	NETHERLANDS	P051130567S	THE SCOTTS COMPANY KENYA LIMITED
31052000	PETERS PROFESSIONAL AND SOLINURE FERTILIZER	17,940	3,434,809	NETHERLANDS	P051130567S	THE SCOTTS COMPANY KENYA LIMITED
31052000	AGROLEAF POWER FERTILIZER	20,790	5,008,456	NETHERLANDS	P051130567S	THE SCOTTS COMPANY KENYA LIMITED
31052000	AGROLEAF POWER FERTILIZER	14,256	3,198,187	NETHERLANDS	P051130567S	THE SCOTTS COMPANY KENYA LIMITED
31052000	AQUA MATRIX FORESTRY FERTILIZER	650	365,720	SOUTH AFRICA	A001237889I	XAMAU JAMES KARURI
31052000	PETERS PROFESSIONAL AND SOLINURE FERTILIZERS	21,900	4,318,702	NETHERLANDS	P051130567S	THE SCOTTS COMPANY KENYA LIMITED
31052000	AGROLEAF POWER FERTILIZER - PART CARGO	14,160	3,284,116	NETHERLANDS	P051130567S	THE SCOTTS COMPANY KENYA LIMITED
31052000	AGROLEAF POWER FERTILIZER High N 31.11.11 TE	40,528	5,998,256	NETHERLANDS	P051130567S	THE SCOTTS COMPANY KENYA LIMITED
31052000	AGROBLEN FERTILIZER	21,000	2,760,713	NETHERLANDS	P051130567S	THE SCOTTS COMPANY KENYA LIMITED
31052000	AGROLEAF POWER FERTILISER - PART CARGO	19,800	3,736,344	NETHERLANDS	P051130567S	THE SCOTTS COMPANY KENYA LIMITED
31052000	AGROLEAF POWER FERTILIZER	14,256	4,024,533	NETHERLANDS	P051130567S	THE SCOTTS COMPANY KENYA LIMITED
31053000	DI-AMMONIUM PHOSPHATE FERTILIZER IN BULK (DAP)	16,400,000	581,847,040	LITHUANIA	P051104351N	YARA EAST AFRICA LIMITED
31054000	2X20FT CONTR STC MONO POTASSIUM PHOSPHATE POWDER(AGRICULTURAL)	42,000	4,514,758	MALAYSIA	P000594739C	ORBIT CHEMICAL INDUSTRIES LTD
31055100	CALCIUM AMMONIUM NITRATE (CAN) FERTILIZER IN BULK PACKED IN 10 LITRE CONTAINERS	9,448,000	236,156,336	SPAIN	P051098567R	MINISTRY OF AGRICULTURE
31059000	MINJINGU ROCK PHOSPHATE FERTILIZER	56,000	786,800	TANZANIA	A003202023T	MUNYI RICHARD GATUMU
31059000	MINJINGU ROCK PHOSPHATE FERTILIZER	30,000	559,999	TANZANIA	A002277261V	MUGANDA ROY NOAH
31059000	AGRICULTURAL CHEMICALS (HYGROCOMBI 25L /FERTILIZERS)	360	8,427	SOUTH AFRICA	P051121532D	HYGROTECH EAST AFRICA LIMITED

Table 17: Aggregated import data by HS Codes (GSS / CountrySTAT Ghana 2010)

PRODUCT \ Indicators	CUSTOM VALUE (GH¢)	NET WEIGHT (KG)
2814100000:Anhydrous ammonia	-	-
3101000000:Animal or vegetable fertilizers	365,139	879,488
3102100000:Urea	604,485	1,000,000
3102210000:Ammonium sulphate	-	-
3102290000:Double salts and mixtures of ammonium sulphate and ammonium nitrate	1,157	2,814
3102300000:Ammonium nitrate	35,751,331	8,794,193
3102400000:Mixtures of ammonium nitrate with inorganic non-fertilizing substances	332,087	258,000
3102500000:Sodium nitrate	850,899	6,893
3102600000:Double salts and mixtures of calcium nitrate and ammonium nitrate	-	-
3102800000:Mixtures of urea and ammonium nitrate in aqueous or ammoniacal solution	-	-
3102900000:Mineral or chemical fertilizers, nitrogenous, nes	8,720	2,000
3103100000:Superphosphates	-	-
3103900000:Mineral or chemical fertilizers, phosphatic, nes	22,417	20,000
3104100000:Carnallite, sylvite and other Crude natural potassium salts	-	-
3104200000:Potassium chloride	463,059	500,000
3104300000:Potassium sulphate	42,888	100,000
3104900000:Mineral or chemical fertilizers, potassic, nes	-	-
3105100000:Fertilizers... in packages of a gross weight ≤10kg	11,754	25,000
3105200000:Mineral or chemical fertilizers with nitrogen, phosphorus and potassium	960,946	1,286,102
3105300000:Diammonium hydrogenorthophosphate (diammonium phosphate)	-	-
3105400000:Ammonium dihydrogenorthophosphate (monoammonium phosphate)	-	-
3105510000:Mineral or chemical fertilizers containing nitrates and phosphates	52,800	6,000
3105590000:Mineral or chemical fertilizers with nitrogen and phosphorus, nes	-	-
3105600000:Mineral or chemical fertilizers with phosphorus and potassium, nes	27,480	1,000
3105900000:Other fertilizers, nes	11,516	4,544

- Data in national currencies / Aggregated by HS codes

From the figures it is clear that we need to record this information in a better or more advanced way. The FAO template is the best way to go.

Table 18: fertilizer domestic availability (FAO)

 FOOD AND AGRICULTURE ORGANIZATION OF THE UNITED NATIONS - STATISTICS DIVISION																		
AGRICULTURAL RESOURCES - FERTILIZER DOMESTIC AVAILABILITY																		
The time reference period is calendar year. Please report data in Metric Tons. If a different unit is used please indicate it, as well as any other relevant note specific to a given data, in the NOTES column. For item descriptions and for other important instructions please refer to the "Explanatory notes" and "Instructions" sheets.																		
HS CODE	COMMODITY	B PRODUCTION (metric tons of product)				C IMPORTS (metric tons of product)				D EXPORTS (metric tons of product)				E DOMESTIC AVAILABILITY (metric tons of product)				NOTES e.g. different year used, different unit used, preliminary data reported, etc.
		2006	2007	2008	2009	2006	2007	2008	2009	2006	2007	2008	2009	2006	2007	2008	2009	
STRAIGHT NITROGENOUS FERTILIZERS																		
2814	Ammonia, anhydrous																	
310230	Ammonium nitrate																	
310221	Ammonium sulphate																	
310240	Calcium ammonium nitrate																	
310210	Urea																	
310280	Urea and ammonium nitrate solutions																	
STRAIGHT PHOSPHATIC FERTILIZERS																		
310310	Superphosphate																	
31031010	Superphosphate above 35%																	
31031090	Superphosphate other																	
2510	Phosphate rock																	
STRAIGHT POTASSIC FERTILIZERS																		
310420	Potassium chloride																	
310430	Potassium sulphate																	
COMPOUND FERTILIZERS																		
310530	Diammonium phosphate (DAP)																	
310540	Monoammonium phosphate (MAP)																	
310551 and 310559	Other NP compounds																	
310551	Other nitrogen & phosphates compounds																	
310559	Other nitrogen & phosphorus compounds																	
310510 and 310520	NPK complex																	
310510	NPK complex <=10kg																	
310520	NPK complex >10kg																	
310560	PK compounds																	
283421	Potassium nitrate																	
OTHER FERTILIZERS (specify below)																		

- Reported by MT of product, by HS class / NPK are aggregated, but can be disaggregated in this form



Fertilizer Trade - Highlights

- Trade data should comprise at least HS codes, volumes, value, country of origin or destination.
- A good description of the fertilizer product is needed to assign the right HS code and to determine the nutrient contents.
- Importation of fertilizers are reported on CIF (cost, insurance & freight) value. Exportation of fertilizers are reported on FOB (free-on-board) value.
- Re-exported fertilizers should not be included on import and export figures of a transit country.
- Trade data shall be reported by fertilizer product.



Quiz

1. Make a distinction between Re-exports and Fertilizer in transit
2. Why are fertilizers in the re-exports excluded in the imports total?
3. A country is importing 200,000 mt of urea, 80,000 mt of DAP and 50,000 mt of NPK 15 15 15. Fill the corresponding FAO production table (table 18)

3.5 Fertilizer consumption

3.5.1 Consumption data from dealers or farmers

Ideally, consumption should relate to the quantities actually used by farmers on crops. This concept is difficult to apply as it requires costly surveys which cannot be conducted regularly. Therefore, it is ruled out as a tool for measuring consumption, on a comprehensive, timely and regular basis.

In general, farmers apply the fertilizers during the year of purchase and sales at retail level can be equated to consumption. In practice, the regular collection of sales data from retail dealers, particularly in large countries, is difficult. Their numbers are large and not all retailers keep a proper record of their sales.

Some countries have developed regulations and legislation concerning packaging, distribution, sale, quality, specifications, price, etc. for fertilizers. The maintenance of records of stocks and sales by the retailer is often included in the regulations. In this case, dealer reporting forms (see table 17) could be sent to the central agency at specified intervals.

3.5.2 Consumption data from manufacturers and sales organizations

In most countries, the consumption statistics are in fact the actual deliveries to agriculture by the manufacturers. The figures may not reflect accurately current consumption in agriculture due to stock changes, but they are reliable since the recording unit is the factory or the trading enterprise (whose numbers are generally small) and such units keep good records of their operations. In the

medium term, the stock changes cancel each other out. The delivery figures are included in the manufacturers monthly report (table 17).

Importers who deliver fertilizers directly to dealers must also send a monthly report to the central agency, giving details on their deliveries by product and by region.

The sum of the sales from two sources (domestic and imported) represents overall consumption, both passing through a common distribution channel.

3.5.3 Apparent consumption

Apparent consumption is used as a proxy for actual consumption. Sometimes it is difficult to do a survey of all the farmers in order to capture actual consumption. We therefore turn to other means that will give us a fair approximate of the actual.

The concept of apparent consumption assumes stability of stock levels at various channels of distribution - private trade, institutional agencies, importing agency (if operating separately). It assumes that supply (availability) equals consumption.



Apparent consumption = Production + Imports – Exports – Non-Fertilizer Use.

When this formula is used to estimate consumption, it implicitly assumes that stocks at the beginning and at the end of the period remain more or less unchanged. In the longer run, stock-level changes should cancel each other out. However, in the shorter term, closing stocks are not normally exactly equal to opening stocks.

This concept has its merits. Production data are normally fairly accurate and reliable since manufacturers keep their records up-to-date and report to the central agency every month. Secondly, the number of manufacturers is normally not very large in any given country. The import and export data are also reliable. The only uncertainties are the opening and closing stocks.

3.5.4 Fertilizer Consumption Tables

Table 19: dealer sales report (FAO)

DEALER SALES REPORT

Name of the dealer.....

Location.....

Period.....

To be submitted by.....

Metric tons			
PRODUCT	NUTRIENT CONTENT % N-P ₂ O ₅ -K ₂ O	QUANTITY (TONS)	PRICE PER TON (including tax)
Product (1)			
Product (2)			
Product (3)			
Product (4)			
Product (5)			

Note that each country may have its own template different from the ones shown. However the final information should be recorded in a FAO template shown below.

Table 20: Fertilizer utilization (FAO)

 FOOD AND AGRICULTURE ORGANIZATION OF THE UNITED NATIONS - STATISTICS DIVISION														
AGRICULTURAL RESOURCES - FERTILIZER UTILIZATION														
HS CODE	COMMODITY	E				F				G				NOTES e.g. different year used, different unit used, preliminary data reported, etc.
		DOMESTIC AVAILABILITY (metric tons of product)				NON FERTILIZER USE (including feed use) (metric tons of products)				USED FOR CROP PRODUCTION (metric tons of product)				
		2005	2006	2007	2008	2005	2006	2007	2008	2005	2006	2007	2008	
STRAIGHT NITROGENOUS FERTILIZERS														
2814	Ammonia, anhydrous													
310230	Ammonium nitrate													
310221	Ammonium sulphate													
310240	Calcium ammonium nitrate													
310210	Urea													
310280	Urea and ammonium nitrate solutions													
STRAIGHT PHOSPHATIC FERTILIZERS														
31031010	Superphosphate above 35%					X	X	X	X					
31031090	Superphosphate other					X	X	X	X					
2510	Phosphate rock													
STRAIGHT POTASSIC FERTILIZERS														
310420	Potassium chloride													
310430	Potassium sulphate													
COMPOUND FERTILIZERS														
310530	Diammonium phosphate (DAP)					X	X	X	X					
310540	Monoammonium phosphate (MAP)					X	X	X	X					
310551 and	Other NP compounds					X	X	X	X					
310510 and	NPK complex					X	X	X	X					
310510 and	NPK blends					X	X	X	X					



Fertilizer Consumption - Highlights

1. Ideally, consumption should relate to the quantities actually used by farmers on crops
2. Getting an accurate value of fertilizer consumption from farmers' usage or dealers' sales is difficult.
3. Apparent consumption is used as a proxy for actual consumption

$$\text{Apparent consumption} = \text{Production} + \text{Imports} - \text{Exports} - \text{Non-Fertilizer Use}.$$
4. Consumption data are reported by fertilizer products (HS).



Quiz

- How can we estimate actual fertilizer use by farmers?
- Use the information below to calculate apparent consumption

Commodity	Production	Imports	Exports	Non-FU	Consumption
Nitrogenous Fertilizers	150 000	50 000	0	0	?
Phosphatic Fertilizers	250 000	0	100 000	5 000	?
Potassic Fertilizers	0	50 000	0	0	?

3.6 Fertilizer consumption in agriculture

3.6.1 Consumption per hectare of crop (fertilizer use by crop)

Not many countries compile fertilizer consumption data by crop. Nevertheless, reliable information on fertilizer use at farm level is very important for a proper planning not only of fertilizer supply and distribution but also, for example, for the planning of cropping and of the research activities and advisory services.

Constraints in collecting Fertilizer Use By Crop (FUBC):

- Statistics on FUBC require technical knowledge and only farmers themselves can provide more accurate information.
- The dealer reporting system mentioned in chapter 5 might be used for estimating the consumption of fertilizers by crops, i.e. the dealer might be asked to assess on which crops the fertilizers he has sold during the period are being used, but this system is unlikely to be practicable where there is a large number of retailers.
- Manufacturers and large dealers often survey their farmer customers for their own planning purposes, but the information is rarely published.
- As a result, therefore, in most situations it will be necessary to carry out farm surveys in order to obtain the required information. In most countries, there is a large number of farms

and it is rarely practicable to carry out a complete census of all of them. A survey of a sample of farmers is used to represent the total.

- Sample surveys also are expensive to carry out and may be feasible only once every three or four years, but they provide very useful information.

The basic information to be sought is the:



- **Area under various crops grown in the year**
- **Proportion of each crop receiving fertilizers**
- **Quantity of fertilizers (product, N, P_2O_5 and K_2O) applied to each crop.**

3.6.2 Consumption per hectare of agricultural land

The average rate of fertilizer consumption of a country can be expressed in terms of a common unit - whether this be the gross cropped area, agricultural land. or arable land of the country - to give nutrient consumption expressed as kg nutrient/ha of the area selected.

Since a few countries have reliable data on fertilizer consumption by crop, inter-country comparisons of consumption on this basis are rarely possible. Under the circumstances, fertilizer consumption per hectare of "agricultural land" is usually used for this purpose. Agricultural land consists of arable land, land under permanent crops and land under permanent pastures. Care needs to be taken when interpreting the figures of countries where substantial quantities of fertilizers are applied to permanent grass.

3.6.3 Fertilizer Use Tables

Table 21: Data collection table FAO for fertilizer use

RATES OF FERTILIZER USE

Season.....

CROP (*)	AREA (Hectares)	AREA FERTILIZED (Hectares)	TOTAL QUANTITY OF NUTRIENTS APPLIED			ESTIMATED RATES OF APPLICATION		
			N kg	P_2O_5 kg	K_2O kg	N kg/ha	P_2O_5 kg/ha	K_2O kg/ha
Crop (1)								
Crop (2)								
Crop (3)								
Crop (4)								

(*) The crops which are important as regards fertilizer use

Countries may find it useful to collect similar information for irrigated and rain fed crops separately. If more than one crop is grown during the year on the same area of land, both the rates of fertilization applied to each crop and the total applied during the year should be recorded.

Table 22: Data collection table IFA - FUBC

Fertilizer Use by Crop Statistics						Color Legend								
Country : _____						Cell to be filled out			Cell to be filled out if information is available			Cell with automatic calculations (Not to be filled out)		
	Planted Area (000 ha)	Average Yield (kg/ha)	Percent of the Planted Area that is Fertilized (%)			Recommended Application Rate (kg nutrient/ha)			Actual Application Rate (kg nutrient/ha)			Total Fertilizer Consumption (000 metric tonnes nutrients)		
			N	P ₂ O ₅	K ₂ O	N	P ₂ O ₅	K ₂ O	N	P ₂ O ₅	K ₂ O	N	P ₂ O ₅	K ₂ O
Crop XYZ (example)														
2006 + 2006/07	2,350.0	5,100	69%	46%	35%	105.0	55.0	50.0	73.0	35.0	20.0	118.4	37.8	16.5
2007 + 2007/08	2,230.0	5,150	71%	47%	37%	105.0	55.0	50.0	75.0	36.0	20.0	118.7	37.7	16.5
Wheat														
2006 + 2006/07												0.0	0.0	0.0
2007 + 2007/08												0.0	0.0	0.0
Rice														
2006 + 2006/07												0.0	0.0	0.0
2007 + 2007/08												0.0	0.0	0.0
Maize														
2006 + 2006/07												0.0	0.0	0.0
2007 + 2007/08												0.0	0.0	0.0
Other Cereals														
2006 + 2006/07												0.0	0.0	0.0
2007 + 2007/08												0.0	0.0	0.0
Soybean														
2006 + 2006/07												0.0	0.0	0.0
2007 + 2007/08												0.0	0.0	0.0
Oil Palm														
2006 + 2006/07												0.0	0.0	0.0
2007 + 2007/08												0.0	0.0	0.0
Other Oilseeds														
2006 + 2006/07												0.0	0.0	0.0
2007 + 2007/08												0.0	0.0	0.0
Cotton														
2006 + 2006/07												0.0	0.0	0.0
2007 + 2007/08												0.0	0.0	0.0
Sugar Crops														
2006 + 2006/07												0.0	0.0	0.0
2007 + 2007/08												0.0	0.0	0.0
Fruits & Veg														
2006 + 2006/07												0.0	0.0	0.0
2007 + 2007/08												0.0	0.0	0.0
Others Crops														
2006 + 2006/07												0.0	0.0	0.0
2007 + 2007/08												0.0	0.0	0.0
TOTAL														
2006 + 2006/07	0.0											0.0	0.0	0.0
2007 + 2007/08	0.0											0.0	0.0	0.0



Fertilizer Use by Crop - Highlights

- The basic information to be sought is the:
 - Area under various crops grown in the year
 - Proportion of each crop receiving fertilizers
 - Quantity of fertilizers (product, N, P_2O_5 and K_2O) applied to each crop.
- Data should be reported for most important crops
- Total fertilizer consumption is calculated per hectare of "agricultural land" i.e. arable land, land under permanent crops and land under permanent pasture



Quiz

1. Which are the three important variables required to compute statistics on fertilizer utilization by a given crop?
2. 50,000 mt of fertilizer NPK 23-10-5 were sold to maize farmers. The recommended rate of application is 100 kg/ha. Total acreage of maize planted this year is 800,000 ha. Calculate the corresponding actual application rate (in nutrient) using Table 21 and 22

3.7 Fertilizer Prices

Ultimately, it is the relationship between crop and fertilizer prices that determines to what extent inputs will be purchased by the farmer and whether the crop price will cover his investment costs and expenses incurred in growing the crops. Sale prices of both the input and the output have to be remunerative for the producers, i.e. the manufacturer and the farmer. In the developing countries a subsidy is often required to keep the input prices low, so that the output prices can be kept at an acceptable level. Experience shows that it is preferable to maintain a reasonable input-output price relationship to enable the farmer to use fertilizer profitably and, at the same time, to keep food grain prices reasonably low. Inputs generally constitute a costly item in the farmers' budget and it may be necessary to subsidize them to achieve the objective of a satisfactory input-output price relationship.

3.7.1 Price definitions

Table 23: price definitions

Ex-factory price (producer price)	At the factory gate for domestically manufactured fertilizer
Ex-port price	At the port of importing country
Ex-warehouse	At the warehouse of the factory or retailer
Wholesale price	At the warehouse of the wholesaler, distributor or sub-distributor. Normally large quantities which are purchased at this price
Retail price	From the dealer at the retail point.
Farm gate price	Price paid by farmer includes the cost of transport to the farm gate.
FAS free alongside ship	The exporter has brought the material to the loading point at the port. Loading onto the ship, ocean freight and other charges are on the importer's account.
FOB free on board	The exporter has loaded the material onto the ship. Ocean freight and other charges are on the importer's account.
CIF cost, insurance and freight.	FOB price + insurance + ocean freight, i.e. the price of the material landed in the importing country. Unloading operations and other expenses at the port of import are on the importer's account

Once the material reaches the importing country, port charges, various other administrative charges, inland railway freight distribution margin, etc. are added to arrive at the sale price at which the farmer purchases the material.

3.7.2 Prices

In developing countries where fertilizer prices are government controlled, they can be obtained without difficulty from the government departments or public agencies supplying the fertilizer.

However, when the fertilizer prices are not government controlled, it is more difficult to obtain reliable price statistics. The information may be obtained from the retailers or from a random sample of farmers. If the level of literacy is low, the information is normally obtained from answers to oral questions put by trained interviewers. In a more developed environment, examination of the accounts or mailed questionnaires may be used.

Normally the collection of information on fertilizer prices will be part of a more general survey covering all farm inputs and possibly other items. It may be reported by dealers or combined with the survey of fertilizer usage. In any event, the standards and methods used must be compatible with existing national, economic and social conditions.

The most important products should be covered and the information requested might include not only the nature of the product, unit weight, price paid and expenses incurred in transporting them to the farm gate, but also the type of pack, the time of the year when the items are bought, etc.

Subsidies, credit at concessional rates of interest and other types of promotional grants are often made available to farmers to enable them to buy the fertilizers. The price reported should be the price they actually pay after deduction of the subsidy. If a tax is added to the price, the tax-inclusive price should be reported.

3.7.3 Price Tables

Table 24: Fertilizer Price Template (FAO)

FERTILIZER PRICES

Source of information.....

Location.....

Prices as on (date).....

Nature of prices.....

PRODUCT	QUANTITY	UNIT WEIGHT	PRICE			TRANSPORT COST TO FARM GATE (if applicable)		
			N kg	P ₂ O ₅ kg	K ₂ O kg	N kg/ha	P ₂ O ₅ kg/ha	K ₂ O kg/ha
Product (1)								
Product (2)								
Product (3)								
Product (4)								

IFDC is collecting since 2010 monthly retail prices for selected fertilizers directly from agro dealers in 20 countries. Detailed and average prices can be accessed in the statistic pages on www.amitsa.org and www.fertilizer.org and monthly publications.

Table 25: IFDC template for monthly price recordings (AMITSA / MIR+)

COMPANY'S CONTACT	
Company name	MENTOL GENERAL SUPPLIERS & AGRO-DEALERS
Town	NTCHEU
Country	MALAWI
Tel	265 235 555
Email	hdzuwa@yahoo.com
Surname	DZUWA
First Name	HELIK DICKSON
Mobile #1	265 991 063 983/993 317 376
Mobile #2	265 888 855 245
Date of collection	9th JANUARY, 2012

Fertilizer type	Fertilizer product	Pack size	Retail price (per pack)	previous month's sales
nitrogen fertilizer [N]	CAN 26 0 0	50 kg bag	7,650.00	10% more
nitrogen fertilizer [N]	UREA 46 0 0	50 kg bag	8,500.00	35% more
complex fertilizers [NPK]	NPK 23 21 0 +4S	50 kg bag	9,500.00	10% less
complex fertilizers [NPK]	NPK 8 18 15 + 6S + 0.1B (D compound)	50 kg bag		
nitrogen fertilizer [N]	CAN 26 0 0	1 kg	250.00	20% more
nitrogen fertilizer [N]	UREA 46 0 0	1 kg	250.00	15% more
complex fertilizers [NPK]	NPK 23 21 0 +4S	1 kg	270.00	40% less

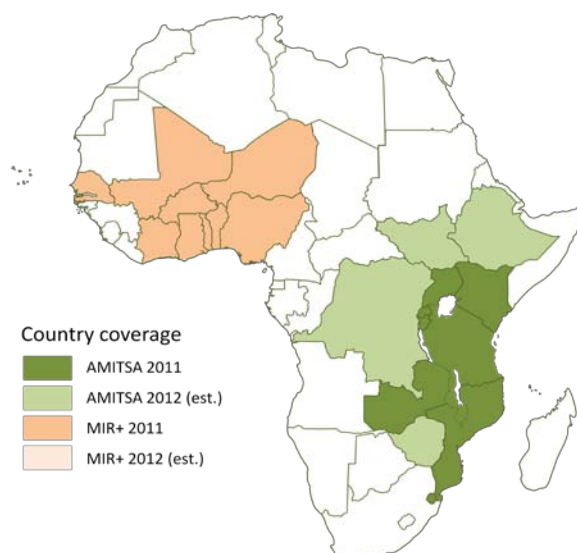
Table 26: IFDC template for monthly national average prices (AMITSA, March 2012)

Selected inputs average retail price (US\$) in selected ESA countries

US\$) per ton	COMPLEX FERTILIZER [NPK]		NITROGEN FERTILIZER [N]		PHOSPHATE FERTILIZER [P]
Country	NPK 17 17 17	NPK 25 5 5 + 5S	CAN 26 0 0	UREA 46 0 0	DAP 18 46 0
Burundi	1090	940		1010	1250
Kenya	900	860	720	830	1050
Malawi			990	1090	
Mozambique			1120	1130	
Rwanda	640	640		800	1010
Swaziland			670	1250	
Tanzania	910	960	860	1000	1170
Uganda	1140	1110	930	1110	1270

Exchange rate (source OANDA time period 02/20/2012 to 03/20/2012) : Burundi=1287.42 BIF/US\$, Kenya=81.57 KES/US\$, Malawi=165.11 MWK/US\$, Mozambique=26.99 MZN/US\$, Rwanda=594.57 RWF/US\$, Swaziland=7.52 SZL/US\$, Tanzania=1570.04 TZS/US\$, Uganda=2384.75 UGX/US\$, Zambia=5183.19 ZMK/US\$

Figure 5: IFDC price collection network in 2011



AMITSA in partnership with COMESA and EAC in Eastern and Southern Africa (10 countries)

MIR+ in partnership with ECOWAS and UEMOA in West Africa (10 countries)

Table 27: Monthly regional prices (SRID / MOFA Ghana, Dec 2010)

**MINISTRY OF FOOD AND AGRICULTURE
STATISTICS RESEARCH AND INFORMATION DIRECTORATE (SRID)
REGIONAL AGRICULTURAL FARM INPUT PRICES-DECEMBER,2010 (IN GH¢)**

COMMODITY	UNIT OF SALE	WESTERN	CENTRAL	G/ACCRA	VOLTA	EASTERN	ASHANTI	NORTHERN	U/EAST	U/WEST	B/AHAFO	AVERAGE
15-15-15	50KG	35.00	27.00	27.00	27.10	40.00	27.00	27.00	40.00	40.00	30.70	32.08
SULP OF AMO.	50KG	40.00	18.00	18.00	19.40	30.00	18.00	18.00	30.00	30.00	21.78	24.32
UREA	50KG	40.00	35.00	25.00	25.00	35.00	25.00	25.00	40.00		29.60	31.07

SOURCE:SRID/MOFA

Table 28: National monthly and annual average prices (SRID/MOFA Ghana, 2010)

NATIONAL AGRICULTURAL FARM INPUT PRICES-YEARLY AVERAGE, 2010 (IN GH¢)

COMMODITY	UNIT OF SALE	JAN	FEB	MAR	APRI	MAY	JUN	JUL	AUG	SEPT	OCT	NOV	DEC	AVE.
15-15-15	50KG	44.4	44.7	48.2	45.2	43.8	41.6	31.1	29.8	29.7	30.1	30.2	32.1	37.6
SULP OF AMO.	50KG	33.1	35.1	33.0	34.0	33.2	30.4	21.5	20.0	20.3	20.4	22.8	24.3	27.3
UREA	50KG	40.5	42.2	43.3	44.0	41.8	37.6	30.1	28.5	28.5	28.8	29.1	31.1	35.5

Having seen various templates from different countries there is the need now to fill in the FAO template since there ought to be uniformity for the purpose of feeding the information on the AfricaFertilizer.org website for public use.

Table 29: fertilizer price paid by farmers (FAO)

 FOOD AND AGRICULTURE ORGANIZATION OF THE UNITED NATIONS - STATISTICS DIVISION										
AGRICULTURAL RESOURCES - FERTILIZER PRICES PAID BY FARMERS										
- The time reference period is calendar year. - Please report data in Local currency per Metric Tons. If a different unit is used please indicate it, as well as any other relevant note specific to a given data, in the NOTES column. - For item descriptions and for other important instructions please refer to the "Explanatory notes" and "Instructions" sheets.										
HS CODE	COMMODITY	I ANNUAL AVERAGE PRICE PAID BY FARMERS (at farm gate) Local currency:				J AMOUNT OF SUBSIDY ☐ Already deducted ☐ Not deducted				NOTES e.g. different year used, different unit used, preliminary data reported, etc.
		2006	2007	2008	2009	2006	2007	2008	2009	
STRAIGHT NITROGENOUS FERTILIZERS										
2814	Ammonia, anhydrous for direct application									
310230	Ammonium nitrate									
310221	Ammonium sulphate									
310240	Calcium ammonium nitrate									
310210	Urea									
310280	Urea and ammonium nitrate solutions									
STRAIGHT PHOSPHATIC FERTILIZERS										
31031010	Superphosphate above 35%									
31031090	Superphosphate other									
2510	Phosphate rock, ground for direct application									
STRAIGHT POTASSIC FERTILIZERS										
310420	Potassium chloride									
310430	Potassium sulphate									
COMPOUND FERTILIZERS										
310530	Diammonium phosphate (DAP)									
310540	Monoammonium phosphate (MAP)									
310551 and 310559	Other NP compounds									
310510 and 310520	NPK complex									
310510 and 310520	NPK blends									
310560	PK compounds									
283421	Potassium nitrate									
OTHER FERTILIZERS (specify below)										



Fertilizer prices - Chapter Highlights

- Retail price is the price paid by farmers from the dealer at the retail point.
- Farm gate price is price paid by farmer, including the cost of transport to the farm gate.
- In case of subsidy, the price reported should be the price farmers actually pay after deduction of the subsidy



Chapter Quiz

- Explain the concept of a subsidy and its knock-on effect on the price chain down to the farmgate price.
- Use data from Ghana 2010 to populate FAO questionnaire on prices (table 29)

NATIONAL AGRICULTURAL FARM INPUT PRICES-YEARLY AVERAGE, 2010 (IN GH¢)

COMMODITY	UNIT OF SALE	JAN	FEB	MAR	APRI	MAY	JUN	JUL	AUG	SEPT	OCT	NOV	DEC	AVE.
15-15-15	50KG	44.4	44.7	48.2	45.2	43.8	41.6	31.1	29.8	29.7	30.1	30.2	32.1	37.6
SULP OF AMO.	50KG	33.1	35.1	33.0	34.0	33.2	30.4	21.5	20.0	20.3	20.4	22.8	24.3	27.3
UREA	50KG	40.5	42.2	43.3	44.0	41.8	37.6	30.1	28.5	28.5	28.8	29.1	31.1	35.5

Chapter 4: Fertilizer “Soft Data” - Policy and Market Development Variables

The understanding of fertilizer statistics requires the explicit understanding of soft data. ‘Soft’ data is a mixture of both quantitative and qualitative data and is collected for policy and market development variables such as policies and regulations, taxes, quality control, fertilizer subsidies, etc. Therefore, this chapter presents the main policy and market development variables, and discusses the best methodologies for estimation and sources for this type of data.



Chapter Learning Objectives

- **Know the different types of ‘soft’ data and understand their role in the development of fertilizer markets in Africa**
- **Understand the current situation in Africa vis-à-vis each of the soft data variables**
- **Know the ‘ideal’ situation with regards to each soft data variable and actions government can take**
- **Learn the best sources of this data**
- **Know how to calculate the indicators using the correct methodology**

4.1 Guidelines for filling the questionnaire

The collection, use and application of soft data will be sourced from administrative and farm survey sources. The official statistics will be based on two main sources – data gathered from statistical surveys, and data extracted from administrative and management data systems.

By using administrative and management data which is already available within administrative or management systems, rather than collecting data afresh, we will limit the overall burden placed on data providers, and also avoid the costs of mounting dedicated data collection exercises. In addition, the information extracted from such systems often has the advantage of being more timely than statistical data.

Such data can be collected from country revenue authorities; statistical agencies; FAO; IFA; NEPAD and CountryStat among other sources. Therefore, for each data requirement we have two possible data sources:

- **Administrative source:** required data may already exist (secondary data) or is available but not yet collected
- **Farm Survey source:** the majority of countries conduct annual farm surveys but in many countries a fertilizer module is not included or a fertilizer module exists but it is not suitable for the calculation of the indicator (s), i.e. does not collect appropriate/relevant data

- For every indicator, include a description of the variable, the data source and methodology used to calculate the indicator (see Tables).
- To facilitate data collection, countries should categorize questions according to data source and assign responsibilities accordingly. So, for example, data and information to be collected through interviews should be grouped together and assigned to Person A; whereas data to be collected by farm surveys should be collated and assigned to Person B.
- Data verification:
 - For each response if an official document exists that substantiates your answer please attach a copy of this document to the completed questionnaire. Such documents would include: copies of the fertilizer law; customs documents showing government importation of fertilizers; newspaper articles, etc.
 - Provide the official response but please also provide any additional documentation/anecdotal evidence from interviews with other sources particularly if the information differs from the official response.

4.2 Resolutions from the Abuja Declaration on Fertilizers use for an African Green Revolution

Note: The following resolutions are excluded:

- *Resolution 11, which refers to the establishment of AFFM. AUC monitors and reports on progress on this resolution*
- *Resolution 12, which calls for NEPAD and the AUC to monitor and report on progress to the AU Heads of State. The questionnaires are filled annually by the countries and on this basis progress reports are compiled by NEPAD and submitted to the African Union Commission.*

4.2.1 Resolution 1 - Increase fertilizer use from 8kg/ha (in 2006) to at least 50kg/ha by 2015

Average levels of fertilizer use in Africa are extremely low (10kg/ha) compared to the world average of over 100 kg/ha. Given the strategic importance of fertilizer in achieving the African Green Revolution to end hunger, the African Union Member States resolve to increase the level of use of fertilizer from the current average of 8 kilograms per hectare to an average of at least 50 kilograms per hectare by 2015.

Table 30: Resolution 1 - Indicators on increasing fertilizer use

Variable	Indicators	Information Required	Data Source	Data Verification
Average level of fertilizer use (per kg/ha of agricultural land)?	_____national average fertilizer use kg/ha	Total apparent consumption of fertilizer in the country Total cultivated land (ha)	Ministry of Agriculture annual farm survey Customs Farmer Organizations	Not necessary

Definitions/explanations

- Cultivated Land: Land under arable and permanent crops
- Agricultural Land: land suitable for agricultural production, both crops and livestock.

4.2.2 Resolution 2: reduce the cost of fertilizer procurement at national and regional levels

Resolution 2: Member States and the Regional Economic Communities should take appropriate measures to reduce the cost of fertilizer procurement at national and regional levels especially through the harmonization of policies and regulations to ensure duty- and tax-free movement across regions, and the development of capacity for quality control. As an immediate measure, member states should eliminate taxes and tariffs on fertilizer and on fertilizer raw materials.

Therefore, Resolution 2 requires that there be a fertilizer policy and regulatory frameworks that are conducive to the development of a sustainable and competitive fertilizer market.

Fertilizer Policy:

Key components of an ideal fertilizer policy are as follows: a) The fertilizer market has been deregulated and liberalized. The private sector is allowed to import and sell fertilizers freely; b) There are no subsidies, price controls, foreign exchange controls, or import quotas for fertilizers; c) The government is not involved in the importation or distribution of fertilizers (for free or at subsidized prices) or the receipt of free or subsidized fertilizers from donors; d) If there are any state-owned enterprises involved in the fertilizer market they do not receive any subventions or other favors. Therefore, the government has established a level playing field for all actors in the fertilizer market. The government's focus is on working to improve the institutional and infrastructural support for the benefit of all market participants.

Table 31: Resolution 2 - Fertilizer Policy

Variable	Indicators	Information Needed	Data Source	Information Verification
Unrestricted private sector participation in fertilizer market	Information (Yes/No)		Ministries of Agriculture, Finance, and Trade	Policy document on private sector participation Importers and agro dealers Development organizations
Fertilizer market is deregulated	Subsidies (Yes/No) Price controls (Yes/No) Foreign exchange controls (Yes/No) Import quotas (Yes/No) Other ____ (Yes/No)	Yes/No	Ministry of Agriculture Ministry of Trade Customs	Documentation on: types of subsidies; controlled price levels; foreign exchange rates Import quotas
Government participation in the fertilizer market	Government importation of fertilizer (by Ministry of Agriculture or SOEs) (Yes/No) Government distribution of free or subsidized fertilizers (Yes/No)	Amount of fertilizer imported by government Amount of fertilizer imported by private sector Amount of free or subsidized fertilizer distributed by government	Ministry of Agriculture Customs	Documentation from the Ministry of Agriculture Importers and agro dealers Development organizations Fertilizer Dealers association

Definitions/explanations

- **Deregulated Fertilizer Market:** This is when government reduces its role in fertilizer market and allows industry greater freedom on how it operates. It entails the removal of government controls from an fertilizer industry to allow for a free and efficient marketplace.
- **Liberalized Fertilizer Market:** Fertilizer market where the prices of fertilizer products are determined by forces of demand and supply. There is less or no government intervention on the determination of prices of fertilizer products.



The effective operations of the fertilizer market require a deregulated environment with minimal government interventions in the market. Governments together with other stakeholders should develop an enabling infrastructure that allows the market to operate freely.



Identify what market interventions by governments distort the free operation of the fertilizer market

Fertilizer Regulatory Frameworks

Countries should develop and implement robust fertilizer regulatory frameworks to ensure that fertilizers of good quality are sold in the market. Countries need to enact a fertilizer law that defines and regulates fertilizer products in the country and develop capacity for effective enforcement of the laws and regulations. Financial resources should be allocated to regulatory agencies and input associations to increase the availability of trained and competent inspectors and analysts to perform quality control functions. The ideal regulatory framework for fertilizer should be comprised of the following: a) compulsory registration of fertilizer manufacturers, importers, and agro dealers; b) Specification and registration of all fertilizers manufactured/imported and sold in the country; c) Guidelines on packaging and labeling on the fertilizer bags (truth-in-labeling); d) Appointment of enforcement agencies and hiring of inspectors; e) Establishment of quality control laboratories and hiring of analysts; f) Prohibition on manufacture/import and sale of non-standard/spurious/adulterated fertilizers. The framework should also have provisions for cancellation of authorization/registration certificates of importers and agro dealers and manufacturers as well as the enforceable punishment for offenders.

Table 32: Resolution 2 - Fertilizer Regulatory Frameworks

Variable	Indicators	Information Required	Data Source	Information Verification
Formal legislation and regulatory framework for fertilizer	Title and date of enactment of Fertilizer Law If no law exists, information on status of draft law and legislation	Fertilizer Law/Legislation	Ministry of Agriculture Interviews agro dealers; regulatory bodies and relevant ministries	Attach fertilizer law if available Relevant regulatory bodies
Inspections of importers and agro dealers premises and testing of fertilizer samples	No of fertilizer inspectors and analysts No of inspections per year No of tests per year	No of fertilizer inspectors and analysts No of inspections per year No of tests per year	Ministry of Agriculture Interviews agro dealers; regulatory bodies and relevant ministries	Importers and agrodealers Regulatory bodies

Definitions/explanations

- **Regulatory Framework:** This is a model people can use to for reforming and enacting regulations in an effective and logical way. Such frameworks are then used to come up with laws and regulations that are flexible and responsive.



- **Fertilizer quality is effectively managed and ensured through regulation that monitors all quality variables.**
- **Regulation ensures compliance as set put in the verification exercises.**
- **Low fertilizer production in Africa with less inter and intra regional trade in fertilizer products.**



1. **Why is it important to ensure that the quality of fertilizer is monitored ?**
2. **What are the major challenges that are faced by countries in ensuring compliance with fertilizer regulatory frameworks ?**

Tariffs and Taxes on Fertilizer

Resolution 2 of the Abuja Declaration 2006 also calls for governments to eliminate tariffs and taxes on Fertilizer. Low fertilizer production in sub-Saharan Africa requires that the fertilizer deficit is met through imports. These imports are mainly from outside Africa. Import tariffs have the effect of increasing the price farmers pay for fertilizers in Africa. Fertilizer is also subjected to taxes which inflate the price. The most typical tax imposed on fertilizers is value-added tax (VAT) which can range anywhere from 15-20%. VAT is typically calculated as a tax rate (%) on the sales value of the fertilizer. Countries may also impose VAT on services and transport of fertilizers and other inexplicable taxes such as withholding tax. Third, customs duty is levied by customs offices on fertilizers, typically in the range of 2.5% of the value of the imported fertilizer. Other taxes include fees charged at the ports and at the borders.

As noted in Chapter 1, there was an unprecedented increase in fertilizer prices in 2007/2008, and although prices have since come down from these unprecedented heights, they have not returned to their pre-2007 levels. Global fertilizer prices continue to be at historically high levels, placing them out of reach of the majority of small-scale farmers in SSA. Reducing import tariffs and customs duties, as well as domestic taxes on fertilizers can lower the price African farmers pay for fertilizers.

Table 33: Resolution 2 - Taxes, tariffs and customs duties on fertilizers

Variable	Indicators	Information	Data Source	Information
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		required		Verification
Import tariffs on fertilizers	Import tariff (Yes/No) Import tariff (%)	Import tariffs	Ministry of Agriculture Ministry of Finance	Supporting documents e.g. customs law Importers
VAT on fertilizer	VAT on fertilizer (Yes/No) VAT (%)	VAT rates	Ministry of Agriculture Ministry of Finance	Supporting documents Importers
Customs duty on fertilizers	Customs duty on fertilizers (Yes/No) Customs duty (%)	Customs duty thresholds Custom duty rates	Ministry of Agriculture Ministry of Finance	Supporting documents Importers
Port charges on fertilizer imports (e.g. IDF Levy)	Port charges on fertilizer imports (Yes/No) Port charges: IDF Levy Other _____	Port charges on fertilizer imports IDF Levy	Ministry of Agriculture Ministry of Finance	Supporting documents Importers Port Authority
Other fees and taxes imposed on fertilizers	Other (Yes/No) Other fees and taxes: Withholding tax (%) Other _____		Ministry of Agriculture Ministry of Finance	Supporting documents Importers

Definitions/explanations

- Import tariffs: Tax levied on the on the value including freight and insurance of imported fertilizer
- Value-Added Tax (VAT): This is a of consumption tax that is placed on a product whenever value is added at a stage of production and at final sale
- Customs Duty: A tax levied on imports (and, sometimes, on exports) by the customs authorities of a country to raise state revenue, and/or to protect domestic industries from more efficient or predatory competitors from abroad. Customs duty is based generally on the value of goods or upon the weight, dimensions.
- IDF Levy: This is a duty called Imports Declaration Fee and is levied on CIF subject to minimum payable.



- Global prices of fertilizers continue to escalate beyond the reach of farmers in SSA.
- Customs duties and import tariffs also add up to the increasing prices of fertilizers leading to marked cost induced reduction in fertilizer use.



State the major reasons why some governments in SSA are reluctant to totally remove import tariffs and duties on fertilizers.

4.2.3 Resolution 3 - Development of agro dealer networks

During the summit the critical role of agro dealers was noted. Resolution 3 of the Abuja declaration 2006 implored countries to formulate policies that deliberately develop agro dealer networks. Agro-dealers are small retailers and stockists who sell fertilizers and other agricultural inputs directly to small-scale farmers.

However, they are typically concentrated in main towns or district headquarters rather than in rural areas close to the farm gate. This makes it difficult for farmers to access inputs. It is common for small-scale farmers to travel 10- 20km just to purchase a bag of fertilizers and at the end of their journey they frequently find that the fertilizer is out of stock.

Moreover, the quantity and quality of agro dealers is limited. The number of agro dealers in a country is typically inadequate to reach the whole of the smallholder farming community. Additionally, marketing and technical skills of stockists are inadequate and they do not have full product knowledge and information about input recommendations. In order to improve the skills of dealers, there is a need for training and technical assistance.

Lack of access to affordable fertilizers has made it virtually impossible for small-scale farmers in Africa to improve their food security and incomes, reinforcing widespread poverty. There is a need to develop a network of skilled and knowledgeable agro dealers located close to the farm gates who make fertilizers available to farmers in a variety of bag sizes, at affordable prices, and in a timely manner.

Table 34: Resolution 3 - Development of agro dealer networks

Variable	Indicators	Information required	Data Source	Information Verification
Private sector share of the fertilizer market	% of fertilizer imported by private sector	Amounts of fertilizer imported last year by government and private sector(	Ministry of Agriculture Ministry of Trade Customs Ministry of Finance	Supporting documentation Importers
Subsidized fertilizer share of the market	% of total fertilizer imported for the government fertilizer subsidy program?	Total fertilizer imported Total fertilizer imported for subsidy program	Ministry of Agriculture Ministry of Trade Agrodealer associations Development projects	Supporting documentation Importers
Spatial coverage of agrodealer network	Number of agrodealers Average distance traveled by farmer to purchase fertilizer	Number of agrodealers Distance travelled by each farmer Number of farmers	Ministry of Agriculture	Supporting documentation Importers
Efficiency of agrodealer network	Average retail price Ratio average retail price/c.i.f price	Prices charged by agrodealers CIF prices	Retailers/agrodealers survey MIS (AMITSA etc.)	Ministry of Agriculture
Effectiveness of agrodealer network	% small-scale farmers using chemical fertilizers	Number of small scale farmers (< 2 hectares) Number of small scale farmers using chemical fertilizer	Ministry of Agriculture annual farm survey	Development partners Importers
Service provision to farmers	Variety of bag sizes sold by agro dealers (50kg, 25kg, 10kg, etc.)	Bag sizes sold bt agro dealers	Interviews/agro dealer survey	Ministry of Agriculture

Definitions/explanations:

- No.1) Total amount of fertilizer imported by the government (No. 2) No.1 minus No. 2) equals: Total amount of fertilizer imported by private sector (No.3)
- % imported by the private sector: the focus here is on the quantity imported by the private sector for its own distribution network. Therefore, quantities imported by the private sector for the government subsidy program should be excluded.



- Agro dealers play a critical role in the easy accessibility of agro-inputs.
- The networks of the dealers are still limited.



Why are agro dealer networks not well developed?

4.2.4 Resolution 4 - develop and strengthen the capacity of the private sector

Resolution 4 calls for governments to take concrete measures to develop and strengthen the capacity of the private sector

The private sector has a critical role to play in the increased supply and use of fertilizers by small-scale farmers in sub-Saharan Africa. In the liberalized African fertilizer market, it is incumbent upon the government to create a conducive business and investment environment to encourage private sector participation and investment in the fertilizer market.

Private sector participation is influenced by factors such as: market size; availability of raw materials for production/blending, macroeconomic environment (exchange rate, interest rate, rural infrastructure); anti-corruption laws; trade regulations; level of political stability; access to finance and technology; and institutions and support services. In many cases, the private sector also lacks the capacity to provide the essential services required by farmers.

The government needs to gain a better understanding of the perceptions, activities and business environment the private sector is operating in, identify the gaps and needs, and on this basis intervene effectively to build the capacity of the private sector to serve the small-scale farming community.

The private sector needs to make available the correct type of fertilizers, at an affordable price, at the right place and in a timely manner. The government can enhance the capacity of the private sector to participate in the fertilizer markets through: investment in rural infrastructure; provision of tax incentives; and promotion of public-private partnerships.

Table 35: Resolution 4 - Capacity building for the private sector

Variable	Indicators	Information required	Data Source	Information Verification
Training courses/workshops by the government and/or development partners	No. of training courses/workshops for private sector per year	No. of trainings by Government No. of trainings by Development partners	Ministry of Agriculture Relevant development partners	Agrodealer association
Tax breaks and/or subsidies (low-interest government loans, etc)	Information and (%)	Low interest government loans Tax breaks Subsidies forms	Ministry of Agriculture Relevant development partners	Agrodealer association
Establishment of public-private partnerships with agri-input businesses	No. of PPPs and details	PPP agreements	Ministry of Agriculture Relevant development partners	Agrodealer association
Other (specify): - _____	Depends on the type of intervention		Ministry of Agriculture Relevant development partners	Agrodealer association

Definitions/explanations

- Tax Breaks: any item which reduces tax, including any tax exemption, tax deduction, or tax credit.
- Public Private Partnerships: describes a government service or private business venture which is funded and operated through a partnership of government and one or more private sector companies. PPP involves a contract between a public sector authority and a private party, in which the private party provides a public service or project and assumes substantial financial, technical and operational risk in the project.



- The private sector plays a critical developmental role in promoting the increased use of fertilizer in SSA.
- The private sector faces capacity gaps and requires Government support for them to better serve the community



1. What are the major reasons why some governments seem not to appreciate the role that is played by the private sector in increasing the use of fertilizers ?
2. State the major capacity constraints that are faced by the private when assisting farmers

4.2.5 Resolution 5 - Improve farmers' access to fertilizer by granting targeted subsidies

Resolution 5: Improve farmers' access to fertilizer by granting targeted subsidies with special attention to poor farmers.

A subsidy is a government policy of making a private good available at below market price. A subsidy usually takes the form of a cash payment to producers/consumers. However, if the government procures goods, such as food grains, at higher than market prices or if it sells at lower than market prices, subsidies are implied. A subsidy can also take many invisible forms: it may be hidden in reduced tax-liability, low-interest government loans, or government equity participation.

Transfers which are straight income supplements need to be distinguished from subsidies. An unconditional transfer to an individual would augment his income and would be distributed over the entire range of his expenditures. A subsidy however refers to a specific good, the relative price of which has been lowered because of the subsidy with a view to changing the consumption/ allocation decisions in favor of the subsidized goods. Even when subsidy is hundred %, i.e. the good is supplied free of cost, it should be distinguished from an income-transfer (of an equivalent amount) which need not be spent exclusively on the subsidized good

Historically, fertilizer subsidies have played a major role in boosting agricultural productivity and food production. Fertilizer subsidies were a central component of the Green Revolutions in Asian and Latin America. In recent years, many African governments have reintroduced fertilizer subsidies in the wake of the call in 2006 by African Heads of State for the introduction of targeted fertilizer subsidies through the Abuja Declaration on Fertilizers for an African Green Revolution; the oil crisis that caused an unprecedented increase in fertilizer prices between January 2007 and mid-2008; and the soaring food prices of 2007/2008 which resulted in many food import-dependent countries in Africa being unable to purchase adequate food stocks to meet their domestic consumption requirements. The emerging demand for biofuels as an energy alternative put further pressure on maize prices and hence fertilizers prices.

The economic justification for fertilizer subsidies is usually one of the following:

- *Efficiency*: fertilizer use by farmers sub-optimal because of lack of information; lack of liquidity; or risk aversion. (value of incremental production should exceed the cost of subsidy)
- *Equity*: transfer income to farmers who are poor or live in remote areas, (or both) and cannot afford to use fertilizers (poor but viable).
- *Externalities*: Fertilizer subsidies justified if fertilizer use generates benefits to others beside the farmer. For example: fertilizer increases plant growth which reduces soil erosion and run-off which then benefits other farmers downstream.

In SSA, subsidies typically take the form of free fertilizers or a subsidy on price. Subsidies on fertilizer in Africa are on average 50% of the market price.

The subsidy can take the form of universal (non-targeted) distribution of free or subsidized fertilizers to the general farming population (typically via government organs) or it can be targeted (via government organs or the private sector). A non-targeted fertilizer subsidy is a subsidy on the price of fertilizer (as illustrated in the table above) and anyone with the money to pay the subsidized price can access the subsidized fertilizer.

The subsidized fertilizer is typically made available by the government organs (it does not use the private sector distribution networks). This type of non-targeted subsidy system has two drawbacks: 1) Farmers who can afford to buy fertilizers at market price can also benefit from the subsidy which leaves less available for farmers who cannot afford to purchase fertilizer at market price, or can only afford to purchase small quantities; 2) It weakens the private sector (“crowding out” effect”) because farmers who can afford to buy fertilizer at market price will instead choose to purchase the subsidized fertilizers, leaving the private sector with unsold stock. The following year, if the subsidy continues, the private sector will import smaller quantities if at all, thus weakening the private sector network.

The Abuja Declaration called for targeted subsidies. A targeted subsidy is aimed at a particular segment of the farming population depending on the objective of the subsidy program.

If the purpose of the subsidy is to boost food production, then the subsidy should be targeted to farmers who already use fertilizers and know how to use them efficiently, but are unable to buy sufficient quantities. Then a subsidy of 25% or 50% will boost their purchasing power so they can afford to buy larger quantities of fertilizers and increase their total output.

If the purpose of the subsidy is to increase food security, then the subsidy should targeted small-scale farmers that are currently non-users as they are too poor to purchase fertilizers.

In either case, the subsidy has to use an instrument to target the desired segment of the farming population. An instrument that has gained popular use is an input voucher. This is a certificate which entitles farmers to buy inputs at a subsidized price. The input vendor (agrodealer) can redeem the voucher for cash from the government.

A targeted subsidy can be administered by government organs or the private sector. In the latter case, the input voucher can be redeemed by farmers at their local agrodealer. This has the advantage of strengthening the linkages between the farming community and the private sector input delivery network and in the case where the objective of the subsidy is food security, the subsidy program will not crowd-out the private sector but rather begin to create linkages with a new customer base.

Table 36: Resolution 5 - Targeted Fertilizer Subsidies

Variable	Indicators	Information Required	Data Source	Information Verification
Fertilizer subsidy	Fertilizer Subsidy (Yes/No)	Fertilizer Subsidy	Ministry of Agriculture	Documentation on fertilizer subsidy from Ministry of Agriculture or other reliable source Development partners Importers and agrodealers
Type of subsidy	Subsidy on price (Yes/No) % subsidy on price Subsidy on transport (Yes/No) % subsidy on transport Subsidy on raw material (Yes/No) % subsidy on raw material Free distribution of fertilizers (Yes/No) Other (specify) (Yes/No)	Type and % of subsidy	Ministry of Agriculture	Development partners Importers and agrodealers
Targeted Subsidy	Input voucher (Yes/No) Fertilizer for work program (Yes/No) Other (specify) (Yes/No)	Type of targeted subsidy	Ministry of Agriculture	Development partners Importers and agrodealers
Mode of distribution of subsidized fertilizer	Distributed by government (Yes/No) Distributed by private sector (Yes/No) Both (Yes/No)	Mode of distribution of subsidized fertilizer	Ministry of Agriculture	Development partners Importers and agrodealers
Timeliness of delivery of subsidized fertilizer	What month did subsidized fertilizer arrive in country? When month does planting season begin?	Month of arrival of subsidized fertilizer Month planting season begins	Customs Ministry of Agriculture	Importers Farmer associations



Targeted Subsidies promote the use of fertilizer in communities in a way that benefits both farmers and the private sector.

Many African governments have reintroduced fertilizer subsidies in the wake of the call in 2006 by African Heads of State for the introduction of targeted fertilizer subsidies through the Abuja Declaration on Fertilizers for an African Green Revolution.



Fertilizer subsidies in some countries have resulted in massive corruption and the beneficiaries abusing the schemes. Some scholars have argued against subsidies saying they distort the market the stakeholders want to freely operate.

Should we promote subsidies? If yes, what needs to be done to ensure that they are put to effective use.

4.2.6 Resolution 6 - Investments in infrastructure

Resolution 6: Accelerate investments in infrastructure, particularly transport, and other measures to improve output market incentives

Demand for fertilizer is derived from the final demand for the crops being produced. Therefore access to output markets is an important determinant of farmer incentives to use fertilizers

Table 37: Resolution 6 - Indicators on investments to improve output market incentives

Variable	Indicators	Information Required	Data Source	Information Verification
Aaverage distance farmers travel (to the nearest district town or wholesale market) to sell their maize (or equivalent staple food crop)?	Aaverage distance traveled _____km	Distance to nearest market from famers Number of farmers who responded	Ministry of Agriculture annual farm survey Agrodealer associations MIS	Ministry of Agriculture
Average transport cost per 50kg bag of fertilizer from a retailer in a large district town to the farm-gate?	Average transport cost per 50kg bag of \$ _____	Number of farmers Average transport cost per 50kg bag of fertilizer from a retailer to farm-gate	Farmer Organizations	Ministry of Agriculture



Investment in infrastructure is very critical for the development of agriculture in Africa. The infrastructural developments include roads and rail networks. This should be supported by fiscal incentives which all lead to support of the output market



List the infrastructure and fiscal incentives that governments can put in place to support output markets

4.2.7 Resolution 7 - Establish national financing facilities for agri-input businesses.

Resolution 7 of Abuja Declaration 2006 calls for member states to establish national financing facilities for agri-input businesses.

Importers and agro dealers in SSA are severely constrained in their activities limited access to finance and liquidity; they often use their own savings to start a business. Some of them get suppliers' credit, but very few of them are able to borrow from commercial or development banks. Resolution 7 of Abuja Declaration 2006 calls for member states to Establish national financing facilities for agri-input businesses. Commercial banks are reluctant to lend to agriculture because of high risk; they impose stringent collateral requirements to minimize the risk of default. While finance is critical to business development, its non-availability adds to transaction costs and restricts the scale of business operations. Few dealers have adequate personal savings to expand the scale of business operations or start a new business.

The following three innovations can assist in removing the financial constraints and improving access to finance for imports and agrodealers in the country.

1. **Credit Guarantee Funds:** The main goal of this fund is to build solid business linkages between importers and agrodealers on the one hand, and commercial banks on the other, by reducing risk for the bank. The arrangement would be as follows: any importer or agrodealer who is interested in starting or expanding the fertilizer business should provide a certain %age (for example, 30%) of the needed capital and the participating commercial bank should provide the balance (for example) 70% of the needed funds, but the commercial bank should get a guarantee for these funds from a fertilizer fund that has been set up by the government. Hence, this arrangement spreads the risk of financing the fertilizer business across importers and agrodealers, commercial banks, and society at large (represented by the government).
2. **Supplier's credit:** Over time and with goodwill, dealers should be linked with wholesalers and importers who could provide 30-60 days credit, thereby reducing the dependence on the fund. Likewise, importers could be linked with global suppliers to get supplier's credit.
3. **Warehouse collateral arrangement** can further reduce the borrowing needs of dealers and importers. A supplier/dealer can bring a large quantity of fertilizer and leave it in a warehouse where the dealer and the commercial bank can have joint ownership, but the bank can release this quantity in small lots so the dealer does not have to arrange for huge sums.

Table 38: Resolution 7- Improving Access to Finance

Variable	Indicators	Information required	Data Source	Information Verification
Public-private initiatives to improve access	PPP initiatives (Yes/No) Number of	PPP agreements	Ministry of Agriculture	Supporting documentation

to finance for importers and agrodealers	initiatives		Importers and agrodealer survey	Importers and agrodealers Development partners
Credit guarantee funds	Credit guarantee funds (Yes/No) Number of funds	Credit guarantee agreements	Ministry of Agriculture Importers and agrodealer survey Interviews with banks	Supporting documentation Importers and agrodealers Development partners
Suppliers' credit	Suppliers' credit (Yes/No) No. of arrangements	Suppliers' credit agreements	Importers and agrodealers survey	Supporting documentation Development partners
Warehouse collateral arrangements	Warehouse collateral arrangements (Yes/No) No. of arrangements	Warehouse collateral arrangements	Importers and agrodealers survey Interviews with banks	Supporting documentation Ministry of Agriculture Development partners



Small businesses in the fertilizer market face funding and liquidity problems. Innovative funding mechanisms should be designed by stakeholders to promote the development of importers and agro dealers.



What are the major reasons why agro dealers and small farmers find it difficult to secure financing from established financial institutions?

4.2.8 Resolution 8 - Regional Fertilizer Procurement and Distribution Facilities

Resolution 8 requires member states to establish Regional Fertilizer Procurement and Distribution Facilities with the support of the African Development Bank, the Economic Commission for Africa, the Regional Economic Communities, and the Regional Development Banks, through strategic public-private partnerships.

Individual country fertilizer purchases are very small and are generally expensive. Fertilizer markets in SSA are typically less than 100,000 tons. As a result, the individual countries cannot realize economies of scale in procurement and transportation, so they end up paying higher prices for imported fertilizers. Regional procurement of fertilizers is a potential solution to this problem. It refers to the joint procurement of fertilizers by importers in two or more countries in a region to realize economies of

scale in procurement, shipping and transportation through bulk purchasing. Ideally, this pooling of import transactions should be private sector based. National governments and development partners can support the initiative by facilitating business linkages between the traders of various countries by providing relevant training, technical assistance, policy workshops and study tours.²

Table 39: Resolution 8 - Regional Fertilizer Procurement

Variable	Indicators	Information required	Data Source	Information Verification
Initiatives to promote regional procurement?	Regional procurement initiatives (Yes/No) Information	Regional procurement agreements	Interviews with importers Ministry of Agriculture	Development partners



Fertilizer trade in Africa should be encouraged. Countries and RECs should quantify and value this trade to improve the knowledge base regarding how much fertilizer trade is already taking place on the continent and how to increase this trade.



State the major constraints on regional procurement

4.2.9 Resolution 9 - promote national and regional fertilizer production and trade

Resolution 9 calls for member states to promote national and regional fertilizer production and intra-regional fertilizer trade to capture a bigger market and take advantage of economies of scale through appropriate measures such as tax incentives and infrastructure development.

Africa has raw materials for fertilizer production. Significant quantities of natural gas are found in Algeria, Angola, Democratic Republic of Congo, Egypt, Equatorial Guinea, Ethiopia, Madagascar, Mozambique, Namibia, Nigeria, Tanzania and Tunisia. There are numerous phosphate rock and potash deposits throughout Africa. However, the deposits of these raw materials in many countries are too small to be commercially viable. For example, only Algeria, Egypt, Togo, Tunisia and Senegal have deposits that are substantial enough to be of economic significance. This is due to unsuitable quantity and/or quality of the raw material, resources that are poorly located in relation to domestic and export markets and/or markets that are too small to realize economies of scale in production.

At the moment, production is concentrated in North Africa and South Africa. Three countries in North Africa, including Egypt and Tunisia, account for 92 % of production. South Africa is the fourth-largest fertilizer producer in Africa. The main fertilizer producers in Sub-Saharan Africa are Mauritius, Senegal and Zimbabwe. Nigeria has also recently resumed production.

Total continental production is 5,791,436 mt of nutrients, which is equivalent to 4 % of world production. South Africa produces phosphate fertilizers, while countries in North Africa, including

² Gregory D.I. and Balu Bumb (2006), "Factors Affecting Supply of Fertilizer in Sub-Saharan Africa" Agriculture and Rural Development Discussion Paper 24, World Bank.

Algeria, Egypt, Libya and Tunisia, produce both phosphate and nitrogen fertilizers. Tanzania, Zambia and Zimbabwe have produced nitrogen and phosphate fertilizers in the past, but production has declined substantially in recent years and current production levels are unknown. However, pre-feasibility studies indicate that production may be economically viable in Angola, Mozambique and Tanzania. Production of phosphate rock for direct application takes place in Burkina Faso, Madagascar, Mali, Senegal and Zimbabwe. Bulk blending plants have been established in Côte d'Ivoire, Malawi, Nigeria, Zambia and Mali.

Table 40: Resolution 9 - National or Regional Fertilizer Production

Variable	Indicators	Information required	Data Source	Data Verification
Do you have any operating fertilizer production plants?	Yes _____ No _____ If Yes, how many? ____	Number of operating fertilizer production plants	Ministry of Agriculture	Importers Manufacturers /blenders
Do you have any operating fertilizer blending plants?	Yes _____ No _____ If Yes, how many? ____	Number of operating fertilizer blending plants	Ministry of Agriculture	Importers Manufacturers /blenders



Low fertilizer production in SSA. This is attributed to capacity and resources constraints



What should be done by the private sector and governments to increase fertilizer production in SSA.

Promotion of inter- and intra-regional trade within and between RECS.

At present, the fertilizer trade that takes place between African countries is very small compared with the amount of fertilizer trade between Africa and the rest of the world. Most fertilizer consumed in Sub-Saharan Africa is imported from outside the continent, and almost all of the phosphate rock and fertilizers produced in Africa are exported outside the continent. Nevertheless, there is some degree of both inter-regional and intra-regional trade of fertilizers in Africa. Much of this trade involves landlocked countries importing from coastal neighbors, but it is also importation from manufacturers in other African countries. It is important to capture these trade flows so that governments can use this information to facilitate increased trade between African countries.

Table 41: Resolution 9 - promote inter- and Intra-REC Fertilizer Trade

Variable	Indicators	Information required	Data Source	Data Verification
Do you import fertilizers from other countries in Africa?	Information (Yes/No) List of countries	Fertilizer imported from other countries in Africa	Importers Use Customs database	Customs
Number of tons and/or %age of total fertilizer imports that are imported from other African countries annually	_____ (tons) _____ (%)	Total fertilizer imported Total fertilizer imported from other African countries	Importers Use Customs database	Customs
Export of nationally produced fertilizers to other African countries?	Information (Yes/No) List of countries	Total fertilizer exported Total fertilizer exported to other African countries	Importers Use Customs database	Customs
Number of tons and/or %age of total fertilizers produced that are exported to other African countries annually?	_____ (tons) _____ (%)	Total fertilizer exported Total fertilizer exported to other African countries	Manufacturers /blenders Use Customs database	Customs



Low fertilizer production in Africa with less inter and intra regional trade in fertilizer products.



Explain why there is limited inter and intra regional trade in fertilizer products

4.2.10 Resolution 10 - Improve farmer access to complementary inputs.

Resolution 10 of the Abuja Declaration 2006 calls on member states to improve farmer access to complementary inputs.

Farmers will only obtain the full benefit of fertilizer use if they use fertilizers in conjunction with hybrid seeds and water. Crop protection products (pesticides, herbicides, and fungicides) are also critical to ensure the crop reaches its full growth potential and for protection of the crop post-harvest. Farmers also need knowledge regarding the correct use of fertilizers and these other inputs and this is best accomplished through technology transfer activities (on-farm trials and field demonstrations).

Market Information is also critical for improving the use and supply of fertilizers. For a well-functioning market, information flows about prices, quantities, stocks, deliveries, and transaction costs should be smooth and timely and should be frequently disseminated—daily, weekly, or monthly. The Ministry of Agriculture in each country should design, establish and operate a market information system (MIS) which provides information on a regular basis. Information about prices, quantities, stocks, deliveries, and other variables of interest should be collected from all important markets and disseminated. The MOA should also monitor the situation with respect to supply, demand, and prices in every district and work with the private sector to alleviate shortages.

Table 42: Resolution 10 - Improved farmer access to complementary inputs

Variable	Indicators	Information Required	Data Source	Information Verification
What %age of small-scale farmers growing maize or the equivalent staple crop, use hybrid seeds?	___%	Total number of small scale farmers growing staple crop Total number of small scale farmers growing staple crop who are using hybrid seeds	Ministry of Agriculture annual farm survey	Supporting documentation Development partners
What %age of small-scale farmers growing maize or the equivalent staple crop, use crop protection products	___%	Total number of small scale farmers growing staple crop Total number of small scale farmers growing staple crop who use crop protection products (pesticides, herbicides, fungicides)	Ministry of Agriculture annual farm survey	Supporting documentation Development partners
What %age of arable land is irrigated?	____% of arable land under cultivation that is irrigated (ha)	Total land cultivated (arable land) in (ha) Total arable land (ha) under irrigation	Ministry of Agriculture annual farm survey Bureau of National or Agricultural Statistics	Supporting documentation Development partners
Has the Ministry of Agriculture established fertilizer trials and field demonstrations during the past year (either alone or in collaboration with NGOs/private sector)?	Yes__No__ No. fertilizer trials ____ No. field demonstrations ____	No. fertilizer trials by government No. field demonstrations by NGOs/Private sector	Ministry of Agriculture	Supporting documentation Development partners
Has the Ministry of Agriculture	Yes__No__	Radio broadcasting of	Ministry of	Supporting

broadcast any radio programs dealing with fertilizers and other agricultural inputs during the past year? (Yes/No)		fertilizer usage programs	Agriculture	documentation Development partners
Has the Ministry of Agriculture broadcast any TV programs dealing with fertilizers and other agricultural inputs during the past? (Yes/No)	Yes___ No___	TV broadcasting of fertilizer usage programs	Ministry of Agriculture	Supporting documentation Development partners
Has the Ministry of Agriculture disseminated any written materials (newsletter, brochures etc.) about fertilizers, correct fertilizer use, and fertilizer markets during the past year? (Yes/No)	Yes___ No___	Written materials dissemination on fertilizer usage programs	Ministry of Agriculture	Supporting documentation Development partners
Has the Ministry of Agriculture collected data on fertilizers and other agricultural inputs during the past year? (Yes/No)	Yes___ No___ —	Fertilizer data collected by the government from previous years	Ministry of Agriculture	Supporting documentation Development partners

Definitions/explanations

- Arable Land: All land under temporary crops (double-cropped areas are counted only once), temporary meadows for mowing or market and kitchen gardens and land temporarily fallow (less than five years).



The full benefits of fertilizer are realized if fertilizer use is complimented with knowledge and correct application of complimentary inputs and services



Agriculture extension services play a critical role in educating farmers and agro dealers. What are the major constraints faced by extension workers?

Chapter 5. Case Studies



Learning Objectives

- Enabling participants to be able to collect data from various sources
- Building familiarity with use of HS codes
- Enabling participants to populate various questionnaires
- Participants should be able to do the validation process using live data
- Building quantitative skills among participants

Case studies are essential and key to understanding the contents of the modules. This affords participants to practice what has been discussed in the modules. The idea is to test whether candidates can recognize various types of fertilizers some of which may be crop-specific and be able to classify them and do the fertilizer analysis

5.1 Converting fertilizer products into nutrients

Fertilizer	Nutrient Content	Material (Tons)	Nutrient		
			N	P ₂ O ₅	K ₂ O
Straight Nitrogenous Fertilizers					
Ammonium Sulphate	21% N	2,153		-	-
Calcium Ammonium Nitrate	26% N	587		-	-
		Sub total=?	?	-	-
Straight phosphatic Fertilizers					
Single superphosphate	?% P ₂ O ₅	1,511	-	272	-
Triple Superphosphate	?% P ₂ O ₅	3,856	-	1,774	-
		Sub total =5,367		2,046	
Straight potassic Fertilizers					
Muriate of potash (KCl)	60% K ₂ O	?			4,920
Sulphate of potash K ₂ SO ₄	50% K ₂ O	?			1,281
		Sub total=10,762			6,201
Complex Fertilizers					
Ammonium Phosphate sulphate	(16-20-0)	5,618	?	?	?
NPK compound	(20-10-10)	6,400	?	?	?
		Sub Total 16,834			
		Grand Total 38,989	5,163	9,556	6,841

(Source: FAO, 1991)



- Populate the table above. You can use the conversion example given earlier in this training module.
- Compute the nutrient ratio for the complex fertilizers in the table and comment

5.2 Collecting import data from Customs data

Fertilizer import data – Kenya (2006 – in kg and KHS)

HS_CODE	PRODUCT DESCRIPTION	QUANTITY	CIF VALUE
31021000	UREA	1	10,154
31021000	UREA 46 PERCENT IN BULK	8,800,000	232,751,600
31023000	AMMONIUM NITRATE	10	70,592
31023000	AMMONIUM NITRATE POROUS PRILLS	20,000	776,419
31026000	CALCIUM NITRATE	50,000	1,331,004
31029000	CALCIUM AMMONIUM NITRATE CAN 26 PERCENT N FERTILIZER IN BUL	15,750,000	292,846,688
31029000	4,400 MTS CALCIUM AMMONIUM NITRATE CAN 26 9 CAO FERTILIZER II	4,400,000	100,818,008
31039000	DIAMMONIUM PHOSPHATE (18-46-0) IN BULK	1,000,000	45,099,148
31039000	DIAMMONIUM PHOSPHATE (18-46-0) IN BULK	5,000,000	225,495,744
31055100	CALCIUM AMMONIUM NITRATE (CAN)FERTILIZER IN BULK PACKED INT	9,448,000	236,156,336
31059000	MINJINGU ROCK PHOSPHATE FERTILIZER	56,000	786,800



- Can you detect some errors or data discrepancies in this table?
- Use these records to generate primary and secondary tables using FAO templates (requires conversions into nutrients)

5.3 Recording imports, converting into nutrients

Fertilizer imports data - Ghana (2006, in old Ghana cedis and kg)

Fertilizer Imports (2006)	CUSTOM VALUE (GHc)	NET WEIGHT (KG)
2814100000:Anhydrous ammonia	-	-
3101000000:Animal or vegetable fertilizers	365,139	879,488
3102100000:Urea	604,485	1,000,000
3102290000:Double salts and mixtures of ammonium sulphate and ammonium nitrate	1,157	2,814
3102300000:Ammonium nitrate	35,751,331	8,794,193
3102400000:Mixtures of ammonium nitrate with inorganic non-fertilizing substances	332,087	258,000
3102500000:Sodium nitrate	850,899	6,893
3102900000:Mineral or chemical fertilizers, nitrogenous , nes	8,720	2,000
3103900000:Mineral or chemical fertilizers, phosphatic, nes	22,417	20,000
3104200000:Potassium chloride	463,059	500,000
3104300000:Potassium sulphate	42,888	100,000
3105100000:Fertilizers... in packages of a gross weight =<10kg	11,754	25,000
3105200000:Mineral or chemical fertilizers with nitrogen, phosphorus and potassium	960,946	1,286,102
3105510000:Mineral or chemical fertilizers containing nitrates and phosphates	52,800	6,000
3105600000:Mineral or chemical fertilizers with phosphorus and potassium, nes	27,480	1,000

Source: MOFA, Ghana



- Use the data to fill in the appropriate table showing fertilizer classes, description and amounts of nutrients in the three important elements of Nitrogen (N), Potash (K₂O) and Phosphorus (P₂O₅)
- Looking at the Ghana Import data, is there any reason to suggest that the data supplied is doubtful and therefore requires validation?
- What do you recommend done to authenticate the validity of this data?

5.4 Describing a Bill of Lading

The Bill of Lading serves as a receipt for goods. It shows evidence of the contract of carriage and document of title to the goods at sea. In the case of road and air transportation the document is known as the Consignment note or Waybill.

Typically the Bill contains both the hard data and soft data. The soft data would be about the name of shipper, the consignee, terms of carriage etcetera and the hard data is the description and quantities of the goods being shipped.

In terms of data collection on exports and imports this would be a good source provided every formal transaction is recorded.

You are supplied with a Bill of lading in the next page for a private importer of fertilizer in Ghana.



- Study the bill and identify the name and type of product being shipped and calculate the nutrient content of N, K and P.
- For the purposes of nutrient calculation which of the two figures would you use, Net mass or Gross mass in kilograms and why?
- A product with 15% N, 15% P and 15% K is written as a 15-15-15 compound. Give the same presentation for the product that was imported by Golden Stork of Ghana.
- Can you explain the importance of the HS code for this consignment?



5182393038377793.3718508621

2 Exporter & Address No.: SCPA SIVEX INTERNATIONAL 83-85 BD VINCENT AURIOL 75013 PARIS FRANCE		1 Declaration Regime 40 3 Gross Mass Kg 111222.00		A Office Code: TMA1 Manifest Ref: 1002591 27/12/2010 BL/AWB: MSCUGK725609A Declaration No: 42010481760 / 0 Date: 29/12/2010	
7 Importer & Address No.: 524M001938 GOLDEN STORK GHANA LIMITED P.O.BOX SC376 TEMA GHANA		8 Consignee / Actual Exporter 524M001938 GOLDEN STORK GHANA LIMITED P.O.BOX SC376 TEMA GHANA		6 Reference number 2220 10121600012TNTLPF02	
9 Declarant/Representative No: TNTLPF02 TIMBER LAND PRODUCTS AND FREIGHT PO BOX 584 TAKORADI GHANA		10 Country of Consig/Destinat TR TURKEY 12 Delivery Terms & Place 06 TEMA		11 Type of Licence 13 Total Invoice Fey 28880.32 CC USD	
14 M Trans./Ship/Aircraft ID, Nationality 3 MSC LARA		15 Total FOB Fey-Import/Export 21215.32		16 Curr Code Rate of Exchange USD 1.461200000	
17 Port of Loading/Unload TRGEM PORT		18 Place Ship/Land & C/Ind TACNK F		19 FOB Fey(Import/Export) 30999.83	
20 Entry/Exit Office CEPS TEMA		21 Identification Warehouse TMA1		22 FCVR No: GLACC0270959TR0002 CRMS Level: 2 FCVR Date: 12/16/2010 12.00	
23 Attached Documents IDF, UCR		24 Term of Payment Bank name: Branch name:		25 Financial and banking data Bank code: A/C No:	
26 Marks & Numbers AS ADD 4 X 20FT CONT STC 2220 BAGS OF AMMONIUM SULPHATE STANDARD AGRICULTURAL FERTILIZER 50KG		27 Item No 001		28 Commodity Code 3102210000	
29 Container Nos: FSCU7604402, MSCU1405967 MSCU2663499, MSCU3286208		30 Qty, Org/Dest TR		31 Zone GEN	
32 Quantity & Unit 2220 BG Description of goods 4 X 20FT CONT STC 2220BAGS OF AMMONIUM SULPHATE - 50KG/BAG (111.08MTMT)		33 Gross Wt Kg 111222.00		34 Net Wt Kg 111075.00	
		35 FOB Fey 30999.83		36 Freight Fey 10712.06	
		37 Other Costs Ney 0.00		38 Supp U1 2220.00	
		39 Licence No: DV		40 Customs Value Fey DQy	
		41 Annexed Docs: SBL, EPL		42 42199.93	

43 Tax	Tax base Amt	TBC	Rate	Amount Exempted	Amount Payable
01	42,199.93	24	0.0000	0.00	0.00
02	42,199.93	30	0.0000	0.00	0.00
03	42,199.93	30	0.0000	0.00	0.00
04	42,199.93	24	0.0000	0.00	0.00
05	42,199.93	24	1.0000	0.00	422.00
06	42,199.93	24	0.5000	0.00	211.00
07	42,199.93	24	0.5000	0.00	211.00
32	30,999.83	25	0.4000	0.00	124.00
33	124.00	33	12.5000	0.00	15.50
48	124.00	33	2.5000	0.00	3.10
Total				0.00	986.60

43 Previous Declaration	Time Limit Mths

Taxes Code	Amount Exempted	Amount Payable
IMP.DUT 01	0.00	0.00
IMP.VAT 02	0.00	0.00
IMP.STX 03	0.00	0.00
IMP.EXC 04	0.00	0.00
PRO.FEE 05	0.00	422.00
ECO.LEV 06	0.00	211.00
EDV.LEV 07	0.00	211.00
NET.CHG32	0.00	124.00
NET.VAT33	0.00	15.50
DGL 42	0.00	422.00
SNF 45	0.00	5.00
NET.NHL48	0.00	3.10
EIDF 72	0.00	5.00

JABBOUR AKKARY

The information and particulars herein entered electronically are true and correct and have been obtained from the original of the documents required for the purpose of this entry.

The information and particulars herein entered are for use by the Ghana Customs, Excise and Preventive Service only.

do hereby declare that :-

14/02/11

11-01-14

Page No : A - 1

5.5 Calculating Fertilizer consumptions and utilizations

Fertilizer consumption in Tanzania (2004/05 to 2008/09, in mt)

TYPE OF FERTILIZER	2004/05	2005/06	2006/07	2007/2008	2008/09
Ammonium Sulfate	2,593	9,386	4,877	4,620	6,672
CAN	12,680	26,166	25,589	12,079	19,070
UREA	54,674	41,835	56,822	69,133	25,846
TSP	2,479	1,185	3,675	140	525
DAP	10,551	14,969	21,438	19,408	9,045
NPK :20:10:10	168	3,704	1,239	6,031	3,982
NPK 25:5:5	7,236	2,570	5,199	3,349	5,225
NPK 10:18:24	18,624	17,332	23,847	23,150	42,244
NPK 4:17:15	-	-	-	-	-
NPK 17:17:17	-	-	126	5,560	1,136
Minjingu rock phosphate(MRP)	-	550	158	587	3,774
MOP/SOP	2,007	1,283	2,237	460	3,744
Others	41	2,020	475	4,972	1,402
Total	111,053	121,000	145,682	149,489	122,665

Source: Ministry of Agriculture , Food Security and Cooperatives (Tanzania)



- Use these consumption figures from to populate FAO fertilizer utilization template for last 2 years

5.6 Computing Fertilizer prices

Monthly national average prices for DAP and urea in selected countries (in US\$)

Fertilizer	Country	Jan-11	Feb-11	Mar-11	Apr-11	May-11	Jun-11	Jul-11	Aug-11	Sep-11	Oct-11	Nov-11	Dec-11
DAP 18 46 0	Kenya	780	821	758	887	857	875	823	798	876	798	974	1,040
	Mozambique				872								
	Tanzania	834	920	901	924	968	950	997	1,117	1,036	1,091	997	1,105
UREA 46 0 0	Kenya	536	606	554	593	643	623	632		666	624	726	788
	Mozambique	919	861	917	762	900	948	1,023		1,123	1,075	1,096	1,175
	Tanzania	594	643	511	672	627	655	704			932	767	954

Source: AMITSA



- Use the Excel function to plot the growth of input prices for Urea and DAP in the 3 selected countries.
- Are there reasonable grounds to explain the increase in fertilizer prices during this year 2010?

Monthly fertilizer prices for Ghana (IFDC, Feb 2012, in new Ghana Cedis)

Product	Unit	Average price (US\$)	Average price (GHC)	Max price in country	Min price in country	RETAIL PRICE OF FERTILIZERS			
						agona swedry	kasoa	techiman	winneba
NPK 15 15 15	50 kg bag	28.0	47.50	55.00	40.00	55.00	45.00	40.00	50.00
NPK 23 10 5 + 4MgO + 2Zn	50 kg bag	23.6	40.00	40.00	40.00			40.00	
Sulfate of Ammonia	50 kg bag	24.6	41.67	55.00	35.00	55.00	35.00	35.00	
Urea 46%	50 kg bag	23.6	40.00	40.00	40.00				40.00

Source: IFDC / MIR+ project



- Use IFDC MIR+ price table for fertilizer in Ghana (February 2012) to populate the FAO monthly price collection template

Annual average fertilizer prices (SRID, Ghana, 2005-2010, in new Ghana Cedis)

Input	Unit	2005	2006	2007	2008	2009	2010
15-15-15	50kg	20.22	20.44	21.72	36.06	43.4	37.57
Sulphate of Ammonia	50kg	15.8	17.54	18.1	28.12	31.69	27.34
Urea	50kg	22.94	24.56	25.82	37.13	41.42	35.47

Source: MOFA, Ghana



- Use SRID (Ghana) data to populate the FAO price questionnaire

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Annexes

Annex 1. Harmonized Codes for Fertilizers

Name of Fertilizer	Description of Fertilizer	HS Codes
Straight Fertilizers		
Ammonia, anhydrous	It is a material mostly produced by the synthetic process and at standard temperature and pressure is a gas. Fertilizer grade anhydrous ammonia contains about 82% of nitrogen.	2814
Ammonium nitrate	It is produced by neutralizing nitric acid (HNO_3) with ammonia (NH_3). Ammonium nitrate may be in white or off white granular or prilled form and coated with a suitable material to prevent absorption of moisture and caking in storage. Pure ammonium nitrate may have a total nitrogen content of about 35%, of which one-half is present as ammoniac nitrogen and the other half as nitrate nitrogen.	310230
Ammonium sulphate	It is produced by reacting ammonia with sulphuric acid (H_2SO_4). It is produced as fine white granules or crystals and contains not less than 20.6% nitrogen in ammoniac form.	310221
Calcium ammonium nitrate	It is produced from ammonium nitrate and finely pulverized calcium carbonate (CaCO_3). It contains not less than 20.5% and up to 28% of nitrogen, half of which is in the form of ammoniac nitrogen and the other half in the form of nitrate nitrogen. It is produced as white, off-white or grey granules or prills	310240
Urea	It is produced from synthetic ammonia and carbon dioxide (CO_2) and contains 46% nitrogen in ammoniac form. Urea may be in granular, prilled or crystalline form.	310210
Urea and ammonium nitrate solutions	They are produced from concentrated solutions of urea and ammonium nitrate by chemical or blending processes.	310280
Superphosphate above 35%	It is produced by the action of sulphuric and phosphoric acids on ground phosphate rock	31031010
Superphosphate other	It is produced by the action of concentrated sulphuric acid on ground phosphate rock	31031090
Superphosphate	It is a fertilizer produced by the action of concentrated sulfuric acid on powdered phosphate rock.	310310
Phosphate rock	It is a natural rock containing one or more calcium phosphate minerals of sufficient purity and quantity as to permit its use directly after grinding or after chemical processing in the manufacture of commercial phosphate fertilizers.	2510

Potassium chloride (Muriate of potash)	It is refined from mined, low-grade naturally occurring ores as the mineral sylvite and in combination with sodium chloride as sylvinite. Potassium chloride contains from about 48 to 62 % K ₂ O.	310420
Potassium sulphate	It is a white crystalline salt and contains 48 to 52 per cent potash (K ₂ O). Potassium sulfate can be extracted from naturally occurring brines or by the decomposition of potassium chloride with sulfuric acid.	310430
Compound Fertilizers		
Diammonium phosphate (DAP)	It is produced by evaporating a solution of phosphoric acid with excess of ammonia.	310530
Monoammonium phosphate (MAP)	It is formed when a solution of phosphoric acid is added to ammonia until the solution is distinctly acid.	310540
Other nitrogen & phosphates compounds	It can be produced as the result of a chemical reaction of nitric acid on phosphate rock, with or without added ammonia and/or phosphoric and/or sulfuric acid or between sulfuric acid and ammonia or by simple mechanical mixing or blending. Other NP compound may also include some AN grades with small amounts of phosphates.	310551
Other nitrogen & phosphorus compounds	As above	310559
NPK complex ≤10kg	It can be produced as the result of a chemical reaction of nitric acid on phosphate rock – the nitrophosphate route, with or without added ammonia and/or phosphoric and/or sulfuric acid or between sulfuric acid and ammonia – the ammoniation route.	310510
NPK complex >10kg	As above	310520
PK compounds	These comprise mixtures of superphosphate or basic slag or ground phosphate rock with straight potash products. PK compounds produced as a result of a chemical reaction are not produced for fertilizer usage.	310560
Potassium nitrate	It can be produced from naturally occurring sodium nitrate and potassium chloride and typically contains 13%N and 45% K ₂ O.	283421
Animal/Vegetable/mixed fertilizer	These are natural fertilizers out of decaying biological matter	310100

Source : FAO , 2012

Annex 2. Fertilizer Data Availability for African Countries

All tables derived from “Collection of Fertilizer Consumption Statistics in Africa” report prepared by Lawrence L. Hammond for The International Fertilizer Industry Association, in preparation of the 2010 IFA Africa Forum, and revised by IFA, FAO and IFDC in June 2011. When countries not listed in tables above are not reporting any data, or are not covered by FAO. More at <http://www.fertilizer.org/ifa/HomePage/INITIATIVES/Africa-Forum>

Availability of export and import data (2011)



Availability of Price and Fertilizer Use data (2011)



Annex 3. **FAO fertilizer questionnaire (2010 / 2011)**

Fertilizer Production

 FOOD AND AGRICULTURE ORGANIZATION OF THE UNITED NATIONS - STATISTICS DIVISION																					
AGRICULTURAL RESOURCES - FERTILIZER PRODUCTION																					
- The time reference period is calendar year. - Please report data in Metric Tons. If a different unit is used please indicate it, as well as any other relevant note specific to a given data, in the NOTES column. - For item descriptions and for other important instructions please refer to the "Explanatory notes" and "Instructions" sheets.																					
HS CODE	COMMODITY	A			B				C = (A/100)*B												NOTES e.g. different year used, different unit used, preliminary data reported, etc.
		Plant nutrient content %			FERTILIZER PRODUCTION (metric tons of product)				FERTILIZER PRODUCTION (metric tons of nutrient content)												
		N	P ₂ O ₅	K ₂ O	2006	2007	2008	2009	2006			2007			2008			2009			
STRAIGHT NITROGENOUS FERTILIZERS																					
2814	Ammonia, anhydrous (82%N will be used if no information is provided)																				
310230	Ammonium nitrate (33%N will be used if no information is provided)																				
310221	Ammonium sulphate (21%N will be used if no information is provided)																				
310240	Calcium ammonium nitrate (26%N will be used if no information is provided)																				
310210	Urea (46%N will be used if no information is provided)																				
310280	Urea and ammonium nitrate solutions (32%N will be used if no information is provided)																				
STRAIGHT PHOSPHATIC FERTILIZERS																					
3.1E+07	Superphosphate above 35% (46%P ₂ O ₅ will be used if no information is provided)																				
3.1E+07	Superphosphate, other (18%P ₂ O ₅ will be used if no information is provided)																				
2510	Phosphate rock (30%P ₂ O ₅ will be used if no information is provided)																				
STRAIGHT POTASSIC FERTILIZERS																					
310420	Potassium chloride (Muriate of potash) (60%K ₂ O will be used if no information is provided)																				
310430	Potassium sulphate (50%K ₂ O will be used if no information is provided)																				
COMPOUND FERTILIZERS																					
310530	Diammonium phosphate (DAP) (18%N and 46% P ₂ O ₅ will be used if no information is provided)																				
310540	Monoammonium phosphate (MAP) (11%N and 52% P ₂ O ₅ will be used if no information is provided)																				
310551 and 310559	Other NP compounds (20%N and 25% P ₂ O ₅ will be used if no information is provided)																				
310510 and 310520	NPK complex (15%N, 15% P ₂ O ₅ and 15% K ₂ O will be used if no information is provided)																				
310510 and 310520	NPK blends (15%N, 15% P ₂ O ₅ and 15% K ₂ O will be used if no information is provided)																				
310560	PK compounds (20% P ₂ O ₅ and 25% K ₂ O will be used if no information is provided)																				
283421	Potassium nitrate (13%N and 45% K ₂ O will be used if no information is provided)																				
OTHER FERTILIZERS (specify below)																					

Fertilizer domestic availability

 FOOD AND AGRICULTURE ORGANIZATION OF THE UNITED NATIONS - STATISTICS DIVISION																					
AGRICULTURAL RESOURCES - FERTILIZER DOMESTIC AVAILABILITY																					
- The time reference period is calendar year. - Please report data in Metric Tons. If a different unit is used please indicate it, as well as any other relevant note specific to a given data, in the 'NOTES' column. - For item descriptions and for ot																					
HS CODE	COMMODITY	A			B				C				D				E				NOTES (e.g. different year used, different unit used, preliminary data reported, etc.)
		Plant nutrient content %			PRODUCTION (metric tons of product)				IMPORTS (metric tons of product)				EXPORTS (metric tons of product)				DOMESTIC AVAILABILITY (metric tons of product)				
		N	P ₂ O ₅	K ₂ O	2007	2008	2009	2010	2007	2008	2009	2010	2007	2008	2009	2010	2007	2008	2009	2010	
STRAIGHT NITROGENOUS FERTILIZERS																					
2814	Ammonia, anhydrous																				
310230	Ammonium nitrate																				
310221	Ammonium sulphate																				
310240	Calcium ammonium nitrate																				
310210	Urea																				
310280	Urea and ammonium nitrate solutions																				
STRAIGHT PHOSPHATIC FERTILIZERS																					
310310	Superphosphate																				
31031010	Superphosphate above 35%																				
31031090	Superphosphate other																				
2510	Phosphate rock																				
STRAIGHT POTASSIC FERTILIZERS																					
310420	Potassium chloride																				
310430	Potassium sulphate																				
COMPOUND FERTILIZERS																					
310530	Diammonium phosphate (DAP)																				
310540	Monoammonium phosphate (MAP)																				
310551 and 310559	Other NP compounds																				
310551	Other nitrogen & phosphates compounds																				
310559	Other nitrogen & phosphorus compounds																				
310510 and 310520	NPK complex																				
310510	NPK complex <=10kg																				
310520	NPK complex >10kg																				
310560	PK compounds																				
283421	Potassium nitrate																				
OTHER FERTILIZERS (specify below)																					

Fertilizer Utilization

 FOOD AND AGRICULTURE ORGANIZATION OF THE UNITED NATIONS - STATISTICS DIVISION																	
AGRICULTURAL RESOURCES - FERTILIZER UTILIZATION																	
- The time reference period is calendar year. - Please report data in Metric Tons. If a different unit is used please indicate it, as well as any other relevant note specific to a given data, in the 'NOTES' column. - For item descriptions and for ot																	
HS CODE	COMMODITY	A			E				F				G				NOTES (e.g. different year used, different unit used, preliminary data reported, etc.)
		Plant nutrient content %			DOMESTIC AVAILABILITY (metric tons of product)				NON FERTILIZER USE (including feed use) (metric tons of products)				USED FOR CROP PRODUCTION (metric tons of product)				
		N	P ₂ O ₅	K ₂ O	2007	2008	2009	2010	2007	2008	2009	2010	2007	2008	2009	2010	
STRAIGHT NITROGENOUS FERTILIZERS																	
2814	Ammonia, anhydrous																
310230	Ammonium nitrate																
310221	Ammonium sulphate																
310240	Calcium ammonium nitrate																
310210	Urea																
310280	Urea and ammonium nitrate solutions																
STRAIGHT PHOSPHATIC FERTILIZERS																	
310310	Superphosphate								X	X	X	X					
31031010	Superphosphate above 35%								X	X	X	X					
31031090	Superphosphate other								X	X	X	X					
2510	Phosphate rock																
STRAIGHT POTASSIC FERTILIZERS																	
310420	Potassium chloride																
310430	Potassium sulphate																
COMPOUND FERTILIZERS																	
310530	Diammonium phosphate (DAP)								X	X	X	X					
310540	Monoammonium phosphate (MAP)								X	X	X	X					
310551 and 310559	Other NP compounds								X	X	X	X					
310510 and 310520	NPK complex								X	X	X	X					
310510 and 310520	NPK blends								X	X	X	X					
310560	PK compounds								X	X	X	X					
283421	Potassium nitrate																
OTHER FERTILIZERS (specify below)																	

Fertilizer Prices (till 2010)

 FOOD AND AGRICULTURE ORGANIZATION OF THE UNITED NATIONS - STATISTICS DIVISION										
AGRICULTURAL RESOURCES - FERTILIZER PRICES PAID BY FARMERS										
- The time reference period is calendar year. - Please report data in Local currency per Metric Tons. If a different unit is used please indicate it, as well as any other relevant note specific to a given data, in the NOTES column. - For item descriptions and for other important instructions please refer to the "Explanatory notes" and "Instructions" sheets.										
HS CODE	COMMODITY	I				J				NOTES e.g. different year used, different unit used, preliminary data reported, etc.
		ANNUAL AVERAGE PRICE PAID BY FARMERS (at farm gate) Local currency:				AMOUNT OF SUBSIDY ☐ Already deducted ☐ Not deducted				
		2006	2007	2008	2009	2006	2007	2008	2009	
STRAIGHT NITROGENOUS FERTILIZERS										
2814	Ammonia, anhydrous for direct application									
310230	Ammonium nitrate									
310221	Ammonium sulphate									
310240	Calcium ammonium nitrate									
310210	Urea									
310280	Urea and ammonium nitrate solutions									
STRAIGHT PHOSPHATIC FERTILIZERS										
31031010	Superphosphate above 35%									
31031090	Superphosphate other									
2510	Phosphate rock, ground for direct application									
STRAIGHT POTASSIC FERTILIZERS										
310420	Potassium chloride									
310430	Potassium sulphate									
COMPOUND FERTILIZERS										
310530	Diammonium phosphate (DAP)									
310540	Monoammonium phosphate (MAP)									
310551 and 310559	Other NP compounds									
310510 and 310520	NPK complex									
310510 and 310520	NPK blends									
310560	PK compounds									
283421	Potassium nitrate									
OTHER FERTILIZERS (specify below)										

Fertilizer – Organic Material (new 2011)

 FOOD AND AGRICULTURE ORGANIZATION OF THE UNITED NATIONS - STATISTICS DIVISION																					
AGRICULTURAL RESOURCES - FERTILIZERS - ORGANIC MATERIAL																					
• The time reference period is calendar year. • Please report data in Metric Tons. If a different unit is used please indicate it, as well as any other relevant note specific to a given data, in the 'NOTES' column. • For item descriptions and for ot																					
HS CODE	COMMODITY	H			I				J				K				L				NOTES (e.g. different year used, different unit used, preliminary data reported, etc.)
		Nutrient content %			PRODUCTION (metric tons of product)				IMPORTS (metric tons of product)				EXPORTS (metric tons of product)				USED FOR CROP PRODUCTION (metric tons of product)				
		N	P ₂ O ₅	K ₂ O	2007	2008	2009	2010	2007	2008	2009	2010	2007	2008	2009	2010	2007	2008	2009	2010	
3101	Organic material, total																				
3101.1	Livestock manure, total																				
3101.2	Cattle manure																				
3101.3	Horse manure																				
3101.4	Pig manure																				
3101.5	Sheep manure																				
3101.6	Poultry manure																				
3101.7	Guano																				
3101.8	Composts																				
3101.9	Sewage waste																				
OTHER ORGANIC MATERIAL (specify below)																					

Annex 4. **NEPAD Country Level Survey**

**Implementation of the Abuja Declaration
on Fertilizers for an African Green Revolution**

Seventh Country Level Survey

This questionnaire should be completed by a Government Official in the Ministry of Agriculture who is in charge of agricultural inputs.

Country: _____ Date: _____

Respondent's details (Please Print):

Name: _____

Title: _____

Ministry/Institution: _____

Email: _____

Tel: _____ Fax: _____

Address: _____

Please return the completed questionnaire as soon as possible to: Dr. Maria Wanzala, Senior Policy Economist, NEPAD Planning and Coordinating Agency. Tel: +27 11 256 3644; Email: mariaw@nepad.org

Table 1. General Questions

Resolutions	Questions (relevant time period is during January 2010 and December 2010)	Answers
Resolution 1. Increase average level of fertilizer use from 8 to 50 kg/ha.	Qu 1. What is the average level of fertilizer use (per kg/ha of cultivated land)?	_____ kg/ha
Resolution 2. Harmonization of Policies and Regulations.	Qu 2a. Does your country have a formal fertilizer law and supporting regulations or do you govern the fertilizer industry by administrative decisions? (tick the one that applies)	Formal fertilizer law and regulatory framework____ Administrative decisions _____
	Qu 2b. Do you carry out fertilizer inspections for quality at the point of sale?	Yes_____ No_____
	Qu 2c. How many fertilizer inspectors do you have for the whole country?	_____ inspectors None _____
	Qu 2d. Do you have tariffs/import taxes on fertilizers?	Yes_____ No_____
	Qu 2e. What other type of taxes do you have on fertilizers? (tick all that apply)	Value-added-tax _____; None_____; Surtax _____; Withholding tax _____; Other (specify) _____
Resolution 3. Develop and Scale-up Dealer Network.	Qu 3a. How many fertilizer importers are there in the country?	_____ importers None _____
	Qu 3b. How many agrodealers are there in the country? (including importer outlets, independent	_____ agrodealers None _____

Resolutions	Questions (relevant time period is during January 2010 and December 2010)	Answers
	dealers, and subsidiaries)	
	Qu 3c. What is the average distance traveled by a small-scale farmer to purchase fertilizer?	_____ km
	Qu 3d. What percentage of farmers uses chemical fertilizers?	_____ %
	Qu 3e. What was the total amount of fertilizer imported for the 2009/2010 agricultural season(s)?	_____ product tons
	Qu 3f. What percentage was imported by the private sector?	_____ %
	Qu 3g. What size of bags is fertilizer sold in? (tick all that apply)	50 kg bags _____; 25 kg bags _____; 10 kg bags _____; Other (specify) _____.
	Question 3h. The purpose of this question is to assess the efficiency of the fertilizer value chain by calculating the ratio of average retail price/c.i.f price. <i>Please refer to Table 2.</i>	
Resolution 4. Capacity building to address fertilizer needs of youth, farmers' associations, civil society organizations, and the private sector.	Qu 4a. Has the Ministry of Agriculture engaged in capacity-building activities for the following groups, either alone or in collaboration with donors, NGOs, etc.? (Yes/No – tick all that apply)	Youth: Yes _____ No _____ Farmers' associations: Yes _____ No _____ Civil society Organizations: Yes _____ No _____ Importers and/or agrodealers: Yes _____ No _____
	Qu 4b. If 4a = Yes, what type of capacity building activities were engaged in? (tick all that apply)	Youth: Training courses _____; Workshops _____; formal education programs _____; Other (specify) _____

Resolutions	Questions (relevant time period is during January 2010 and December 2010)	Answers
		<i>Farmers' associations:</i> Training courses ____; Workshops ____; formal education programs ____; Other (specify) ____ <i>Civil society organizations:</i> Training courses ____; Workshops ____; formal education programs ____; Other (specify) ____ <i>Importers and/or agrodealers:</i> Training courses ____; Workshops ____; formal education programs ____; Other (specify) ____
Resolution 5. Granting of targeted fertilizer subsidies, with special attention to poor farmers.	Qu 5a. Does your country have a subsidy on fertilizer? (Yes/No).	Yes ____ No ____]
	Qu 5b. If 5a = Yes, what type of subsidy? (tick all that apply)	Subsidy on price ____; Subsidy on transport ____; Subsidy on raw material ____; free distribution of fertilizers ____; other (specify) ____
	Qu 5c. If 5b=Yes, what type of instruments are used to target the subsidy? (tick those that apply)	Input vouchers: ____; fertilizer for work programs: ____; other (specify): ____
	Qu 5d. Is there any importation of fertilizers by governments/donors/NGOs?(Yes/No)	Yes ____ No ____
	Qu 5e. If 5c = Yes, is the fertilizer imported by government/donors/NGOs distributed by governments/donors/NGOs or via the private sector? (tick all that apply)	Distributed by government/donors/ NGOs ____ Distributed by private sector ____ Both ____

Resolutions	Questions (relevant time period is during January 2010 and December 2010)	Answers
Resolution 6. Accelerating infrastructure investment to improve output market incentives.	Qu 6a. How many kilometers of all-weather roads are there in the country?	Kms of all-weather roads: _____kms
	Qu 6b. How many kilometers of functioning railway lines are there in the country?	Kms of functioning railway lines: _____kms
Resolution 7. Establish National Financing Facilities for Input Suppliers.	Qu 7a. Are there any public, private, and/or public-private initiatives to lower risks to financial institutions of lending to importers/agro-dealers (Yes/No)	Yes _____ No _____
	Qu 7b. If 7a = yes, what type of initiatives (tick all that apply)	Credit guarantee funds _____; government sponsored programs (e.g. subsidized interest rates) _____; donor sponsored programs _____; Other (specify) _____
Resolution 8: Establish regional fertilizer procurement and distribution centers	Regional procurement of fertilizers refers to the joint procurement of fertilizers by importers in two or more countries in a region to realize economies of scale in procurement, shipping and transportation. Through bulk purchasing. Please briefly express your views regarding the potential advantages and disadvantages of this practice.	<i>Advantages:</i> _____ _____ <i>Disadvantages:</i> _____ _____ _____
Resolution 9. Promote national/regional fertilizer production and intra-regional trade.	Qu 9a. Do you have any operating fertilizer production plants? (Yes/No)	Yes _____ No _____

Resolutions	Questions (relevant time period is during January 2010 and December 2010)	Answers
	Qu 9b. Do you have any operating fertilizer blending plants? (Yes/No)	Yes_____No_____
	Qu 9c. Do you import fertilizers from other countries in Africa? (Yes/No)	Yes_____No_____
	If 9d = Yes, please list the countries	
	Qu 9e. Do you export fertilizers to other countries in Africa? (Yes/No)	Yes_____No_____
	If 9f = Yes, please list the countries	
Resolution 10. Improve farmer access to complementary inputs (quality seeds, irrigation facilities, extension services, market information).	Qu 10a.What percentage of farmers uses hybrid seeds?	_____%
	Qu 10b. What percentage of farmers uses crop protection products?	_____%
	Qu 10c. What percentage of arable land is irrigated?	% of arable land irrigated (ha); _____
	Qu 10d. What types of irrigation investments have been made? (tick all that apply)	Treadle pumps ____; Drip irrigation ____; Canals ____; Bunds ____; Terraces ____; Farm ponds ____; Rehabilitation of large irrigation schemes ____; Development of large irrigation schemes _____.
	Qu 10e. Has the Ministry of Agriculture established fertilizer trials and field demonstrations during January 2008 and June 2008 (either alone or in collaboration with NGOs/private sector)? (Yes/No)	Yes_____No_____
	Qu 10f. If 10c = Yes, How many fertilizer trials and field	_____Fertilizer trials

Resolutions	Questions (relevant time period is during January 2010 and December 2010)	Answers
	demonstrations were established during January 2010 and December 2010?	_____ Field demonstrations
	Qu 10g. Has the Ministry of Agriculture broadcast any radio programs dealing with fertilizers and other agricultural inputs during January 2010 and December 2010? (Yes/No)	Yes _____ No _____
	Qu 10h. Has the Ministry of Agriculture broadcast any TV programs dealing with fertilizers and other agricultural inputs during January 2010 and December 2010? (Yes/No)	Yes _____ No _____
	Qu 10i. Has the Ministry of Agriculture disseminated any written materials (newsletter, brochures etc.) about fertilizers, correct fertilizer use, and fertilizer markets during January 2010 and December 2010? (Yes/No)	Yes _____ No _____
	Qu. 10j. Has the Ministry of Agriculture collected data on fertilizers and other agricultural inputs during January 2010 and December 2010? (Yes/No)	Yes _____ No _____

Note: Two resolutions have not been included in this questionnaire: Resolution 11 (Establish the Africa Fertilizer financing Mechanism) is excluded as it refers to actions to be taken at the continental level by the African Development Bank with support from the African Union/NEPAD and Economic Commission for Africa; and Resolution 12 (Establish a monitoring and evaluation tool and provide bi-annual reports to the African Union) as this refers to action to be taken by the AU and NEPAD.

Table 2. Data for the calculation of the ratio of the average retail price to the c.i.f price (Question 3h)

	Status as of December 2010	
Top 4 fertilizer products in country	Average retail price per US\$/50kg bag ¹	Average c.i.f price for each fertilizer product US\$/50kg bag ¹
Product 1:		
Product 2:		
Product 3:		
Product 4:		

¹For each fertilizer product, ask the 4 largest importers for their prices and take the average.

Table 3. Existing or Ongoing Fertilizer Initiatives

In the space below please describe any existing measures and interventions implemented to increase fertilizer use and supply (goals, objectives, key activities involved, and stakeholders) and describe and quantify measures needed to scale-up or improve existing initiatives to increase fertilizer use and supply. This should represent your flagship program for the implementation of the Abuja Declaration on Fertilizers. It may be in areas such as: developing a network of rural input dealers, developing linkages between input and output markets to improve incentives to use fertilizer, private sector development, local manufacturing or blending, improving access to finance for importers and dealers, and legal and regulatory framework

Annex 5. Fertilizer Statistics Country Digest

Timeline

Here, insert the calendar and periodicity for data collection [C], processing [P] and validation [V]

Data	Periodicity	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Production													
Import / export													
Prices (retail)													
Prices (FOB, CFR)													
Consumption													
Use by crop, by ha													

Preferred sources

Data	Preferred source	Data accuracy (scale: 0-5)
Production		
Import / export		
Prices (retail)		
Prices (FOB, CFR)		
Consumption		
Use by crop, by ha		

Validation

Data	Periodicity	Validation by	Most critical checks
Production			
Import / export			
Prices (retail)			
Prices (FOB, CFR)			
Consumption			
Use by crop, by ha			

Annex 6. **AFO Fertilizer Catalogue format (sample)**

Country	Fertilizer Category	HS Code	Fertilizer Product	Brand	N	P	K	Mgo	S	B	Zn	Fe	Major Crops Usage	Recommended Application Rate
Ghana	Phosphate Fertilizer [P]	3103	DAP 18 46 0		18	46	0							
Ghana	Complex Fertilizer [NPK]	31055000	NPK 15 15 15		15	15	15						on maize and wheat at planting	100 kg/ha on maize 200 kg/ha on wheat
Ghana	Nitrogen Fertilizer [N]	3102	UREA		46	0	0						Top Dressing	100 kg/ha on maize 200 kg/ha on wheat
Ghana	Nitrogen Fertilizer [N]	310221	Ammonium Sulfate	YaraBela Sulfan	24	0	0	6						